机械与电子工程学院
College of Mechanical and Electronic Engineering

第八章 组合变形

Chapter 8 Combined deformation

刘洪萍

内容

§8-1 组合变形和叠加原理

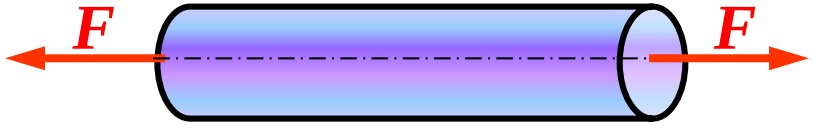
（Combined deformation and superposition method）

§8-2 拉伸（压缩）与弯曲的组合

（Combined axial and flexural loads）

§8-4 扭转与弯曲的组合

（Combined torque and flexural loads）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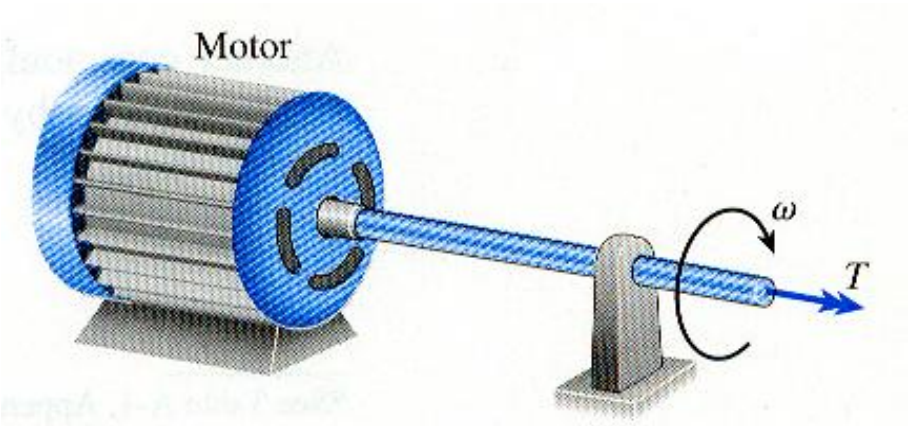
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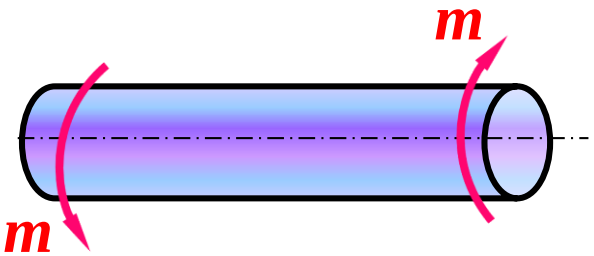
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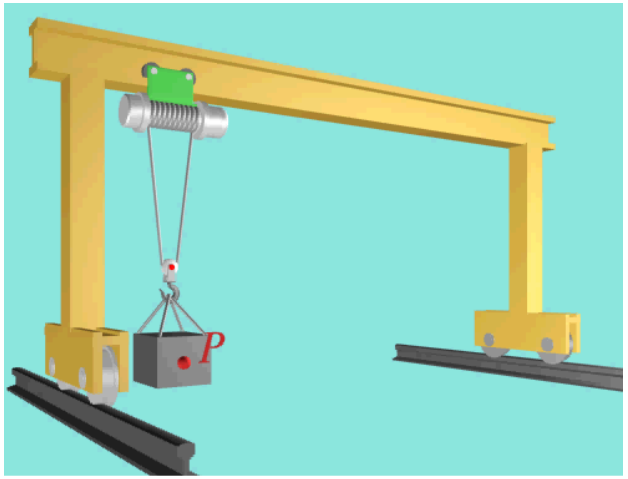
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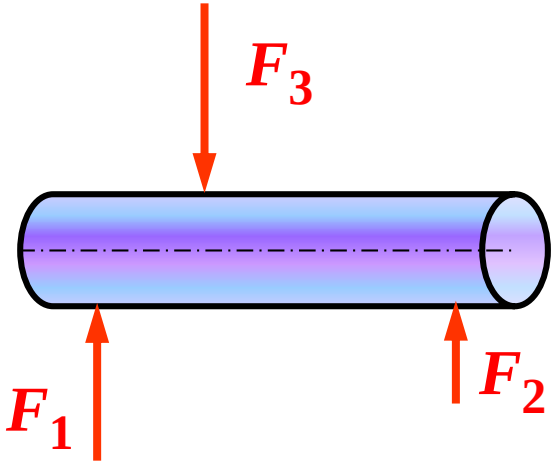
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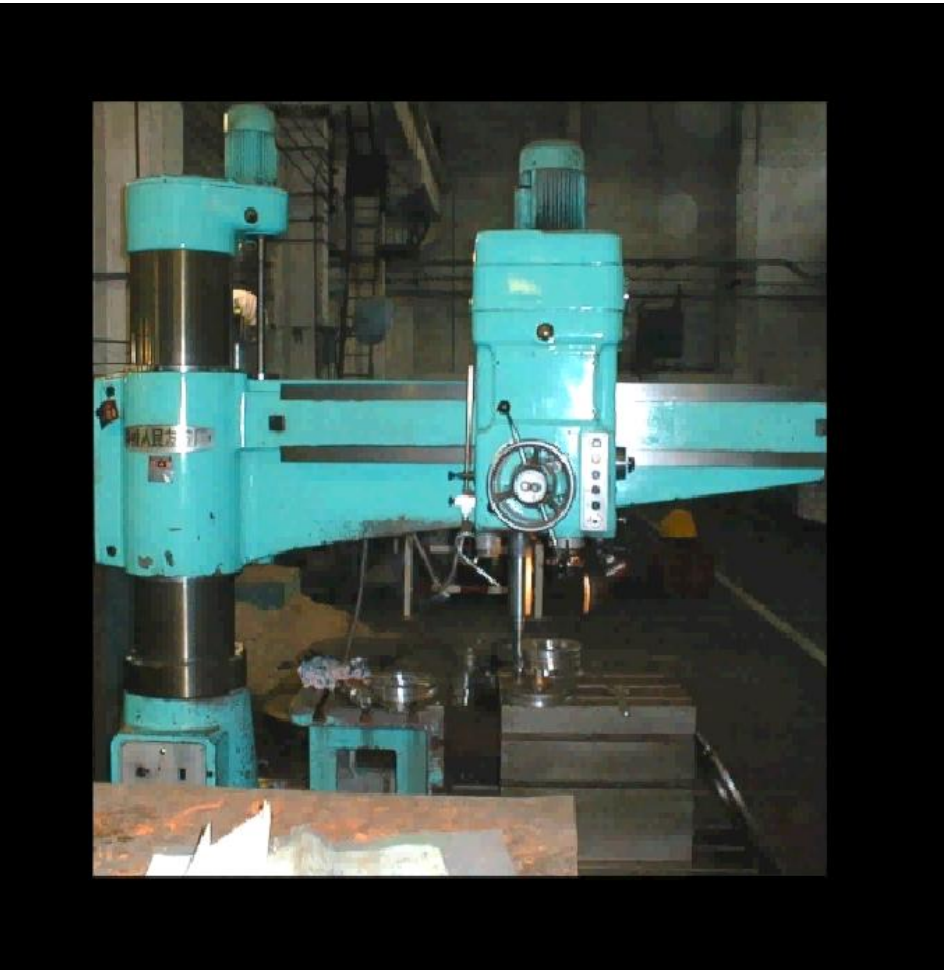
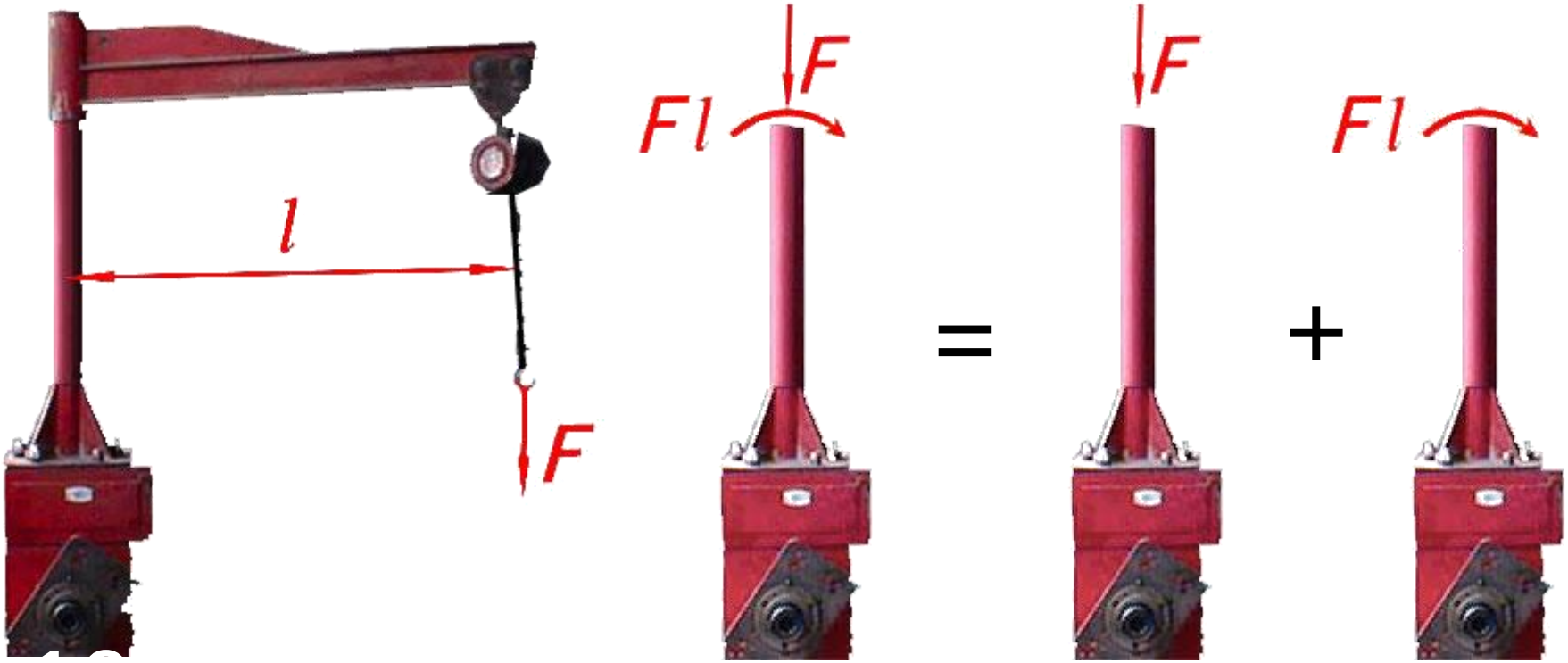
§8-1 组合变形和叠加原理

Combined deformation and superposition method

一、组合变形的概念（Concepts of combined deformation）

构件在荷载作用下发生两种或两种以上的基本变形，则构件的变形称为**组合变形**。

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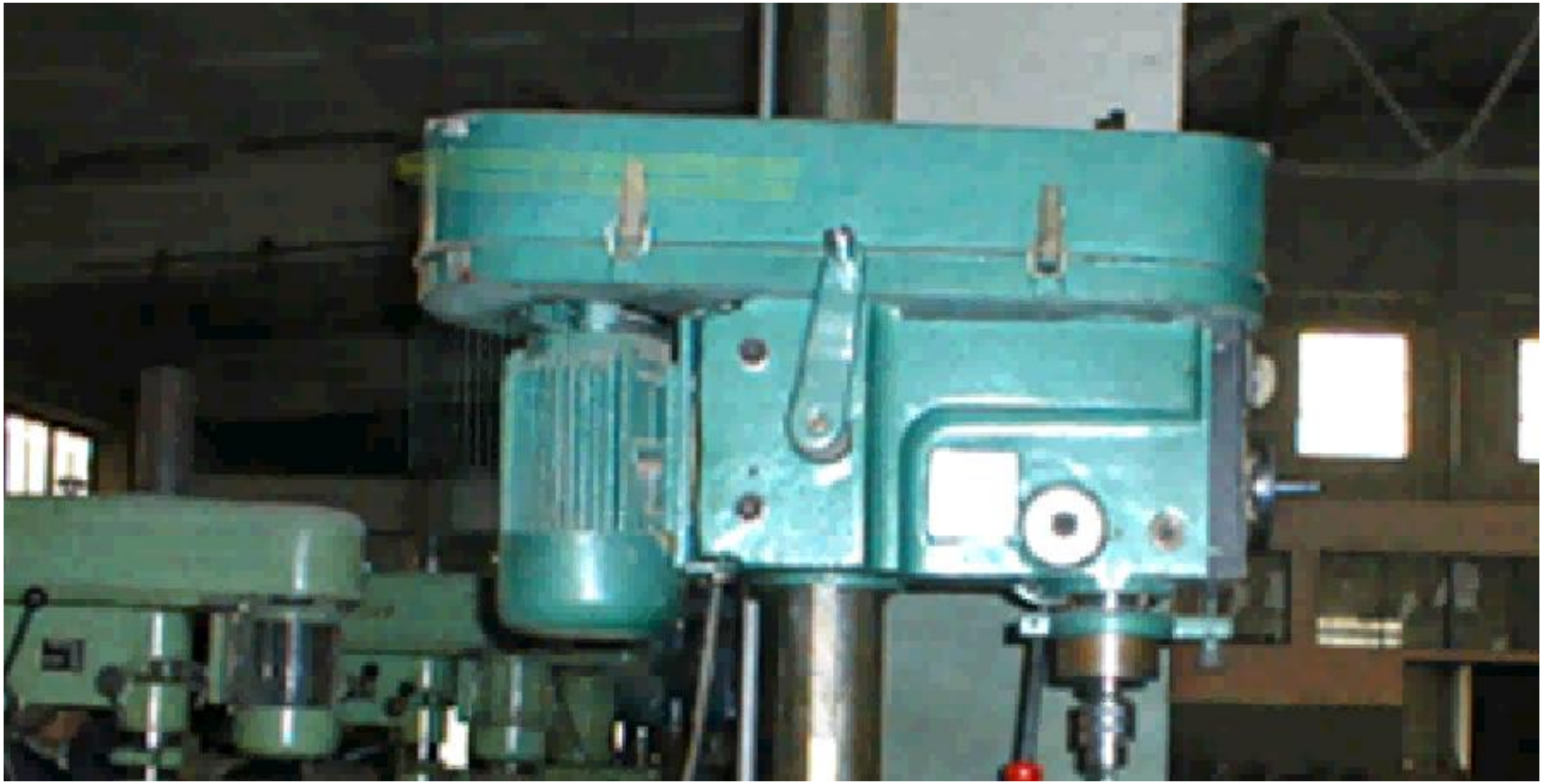

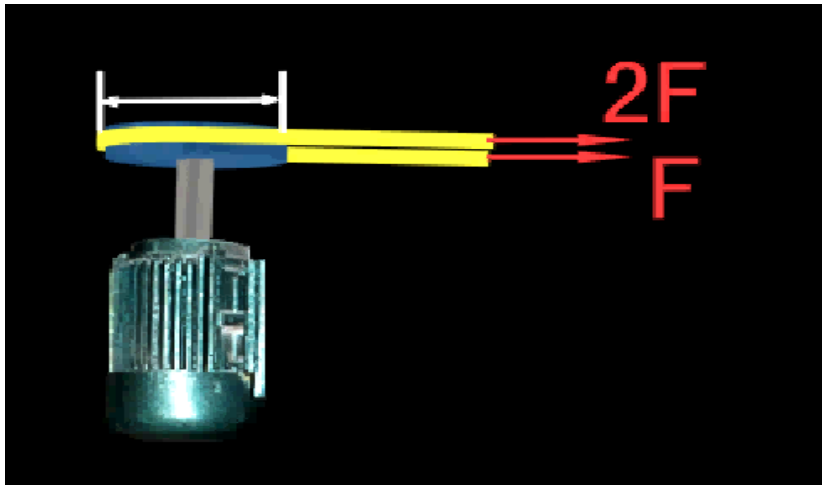
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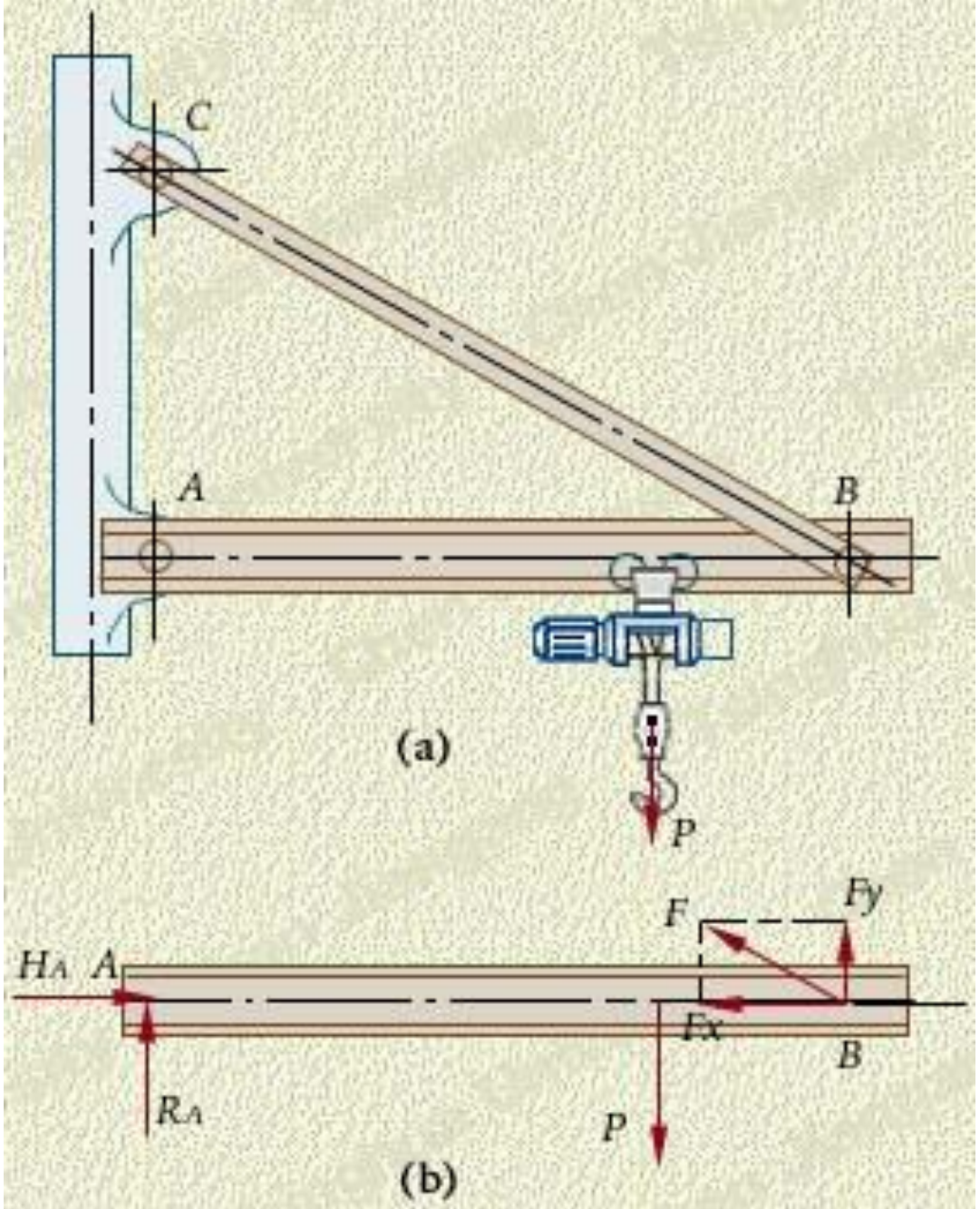


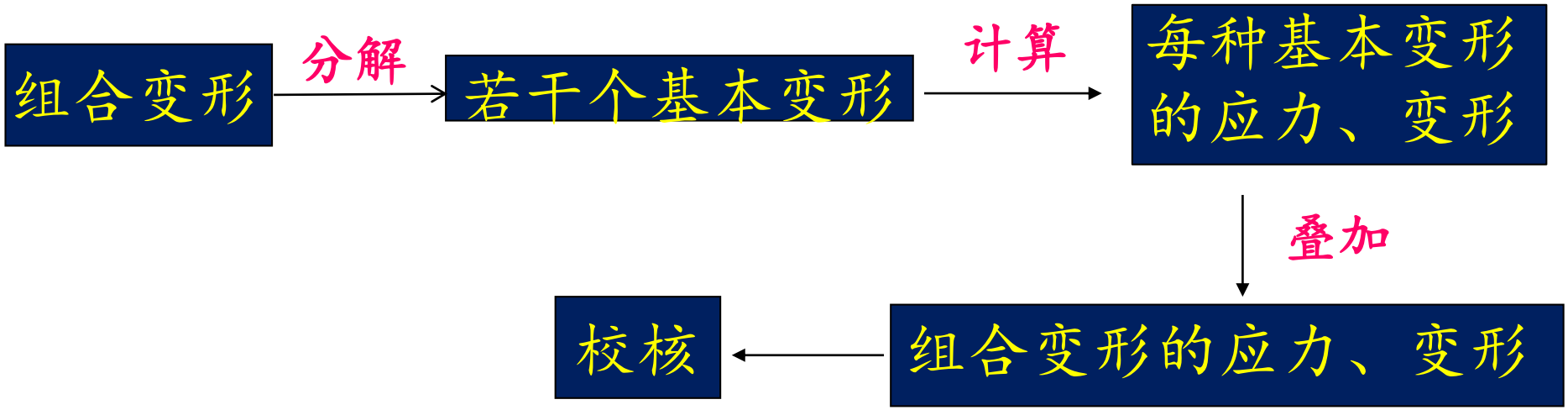
起重机横梁的受力简图

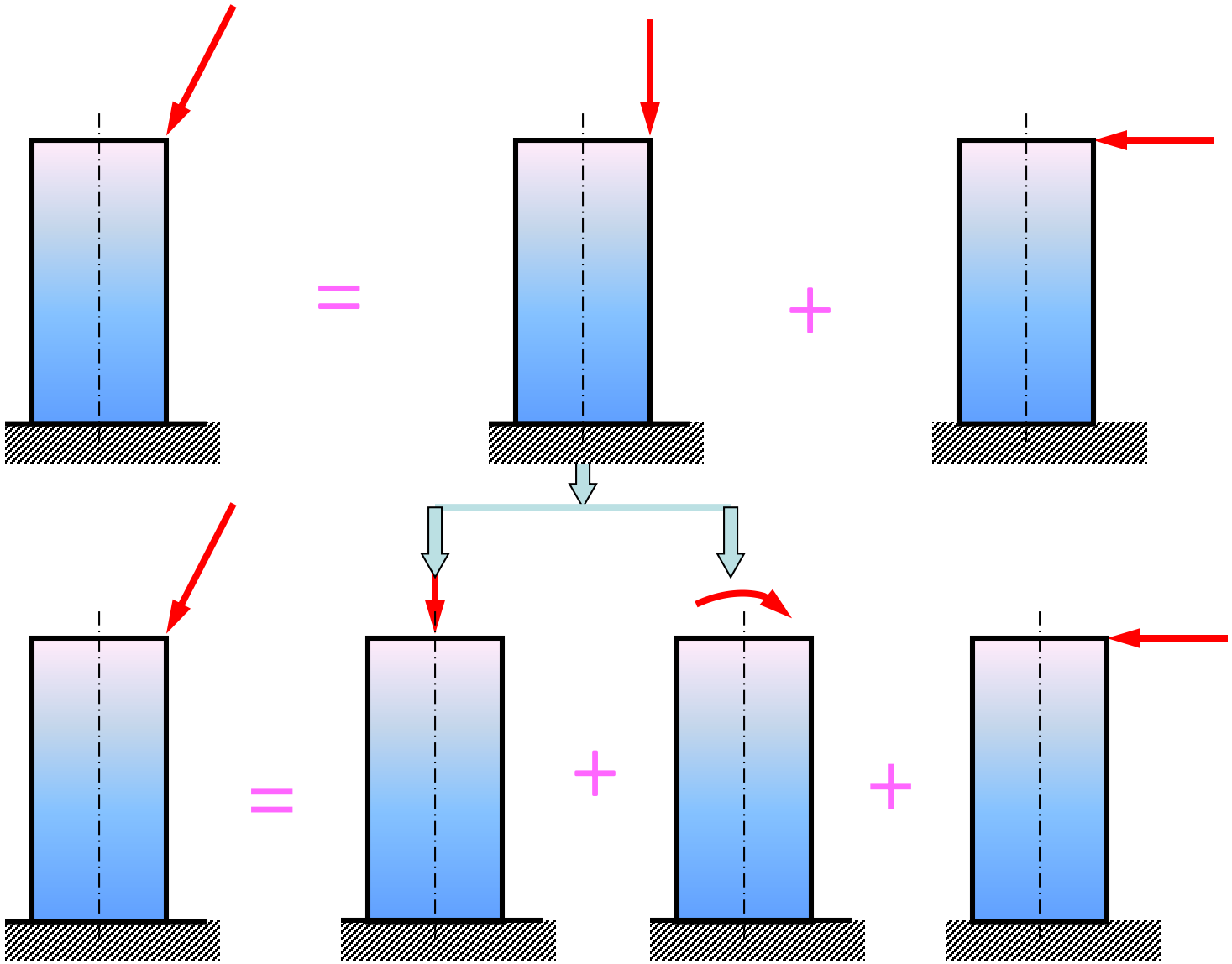
二、解决组合变形问题的基本方法—叠加法

Basic method for sloving combined deformation-superposition method

前提：线弹性，小变形

叠加原理的成立要求：内力、应力、应变、变形等与外力之间成线性关系。





三、研究内容

组合变形的强度问题。

变形限于拉压与弯曲的组合，弯扭组合。

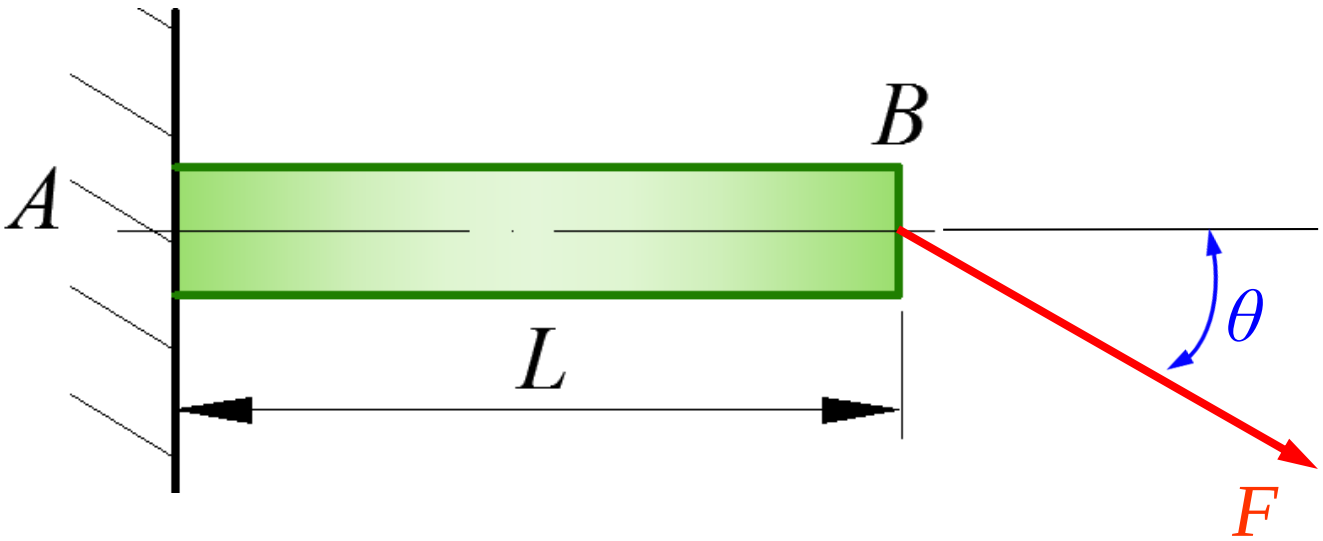
无新知识点，前面各章内容的组合。

四、地位

强度问题的研究，基本变形的研究是基础，应力状态分析是纵深发展，组合变形是横向发展，强度理论形成一完整体系。

§8-2 拉伸（或压缩）与弯曲的组合

（Combined axial loading and bending）



F位于纵向对称面内，作用线过B段形心，与轴线夹角为θ。杆长为L。

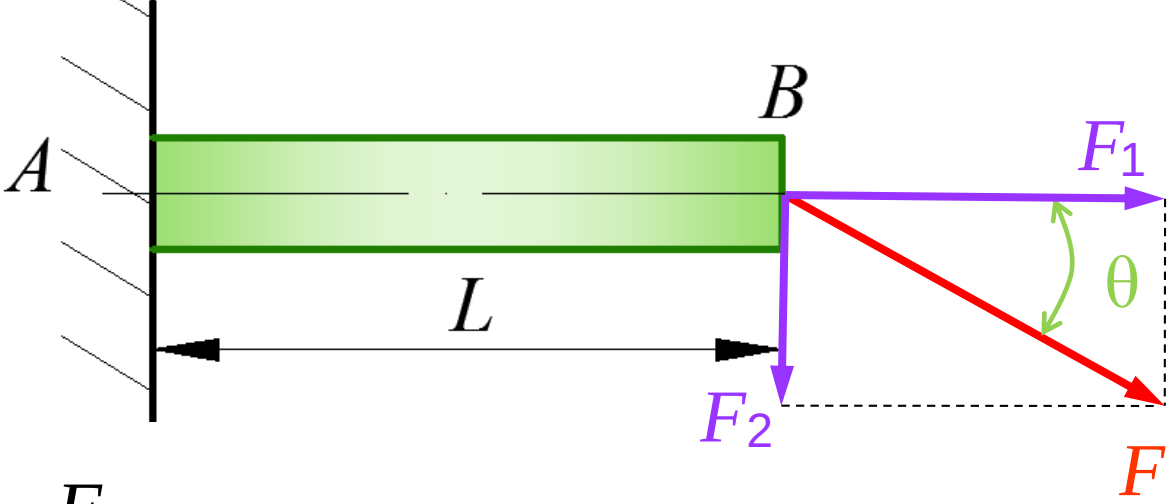
一、外力分析

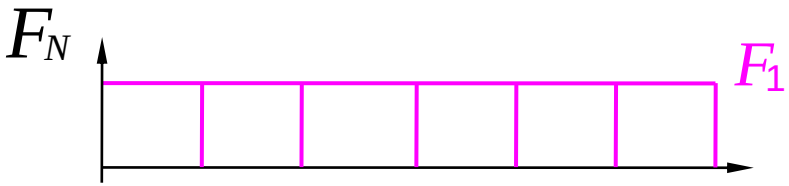
目的：变形形式

$F_1=F\cos\theta$ 拉伸变形

$F_2=F\sin\theta$ 弯曲变形

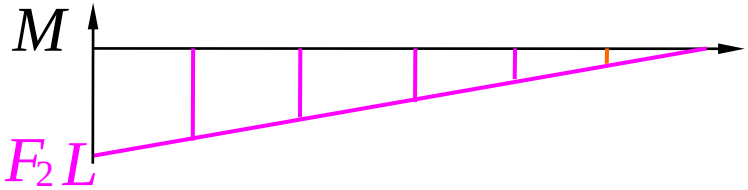
结论：拉伸弯曲
组合变形





二、内力分析

目的：危险截面



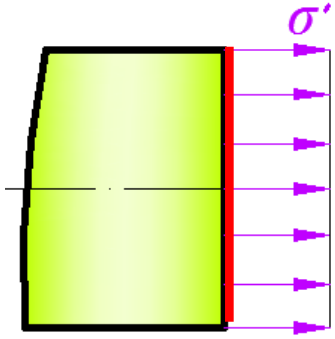
剪力的影响不计。

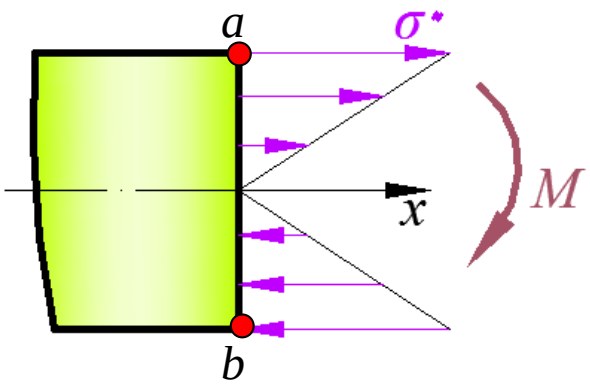
结论：A为危险截面。

三、应力分析

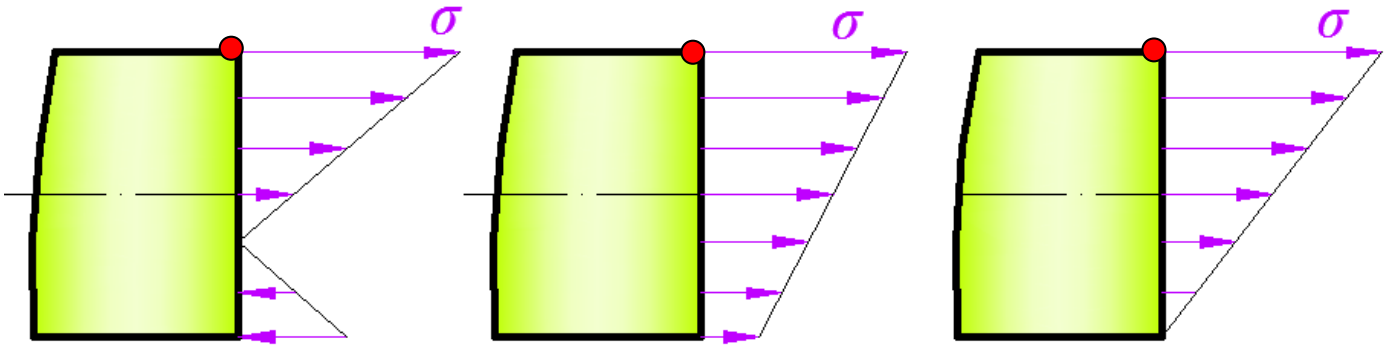
目的：危险点

$$\sigma' = \frac{F_1}{A}$$

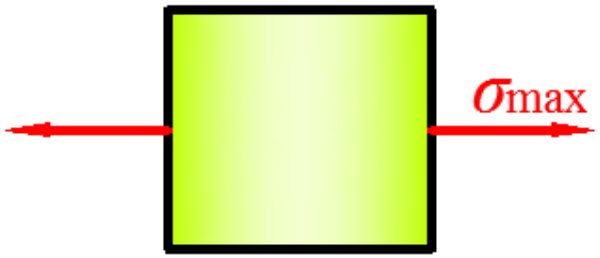




$$\sigma'' = \frac{F_2 L}{W_z}$$



$$\sigma_{\max} = \frac{F_2 L}{W_z} + \frac{F_1}{A} \qquad \sigma_{\min} = -\frac{F_2 L}{W_z} + \frac{F_1}{A}$$



结论：最大应力在a点，最小应力在b点。单向应力状态。

四、强度条件

$$\sigma_{max} \le [\sigma]$$

当材料的许用拉应力和许用压应力不相等时,应分别建立杆件的抗拉和抗压强度条件。

$$\sigma_{tmax} \le [\sigma_t]$$

$$\sigma_{cmax} \le [\sigma_c]$$

解组合变形的一般步骤

外力分析—绘受力简图

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内力分析—找危险面

↓

应力分析—找危险点

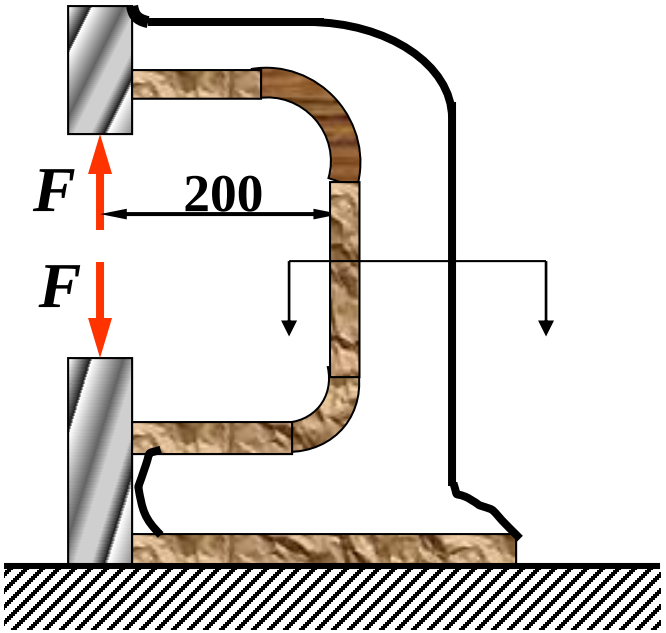
↓

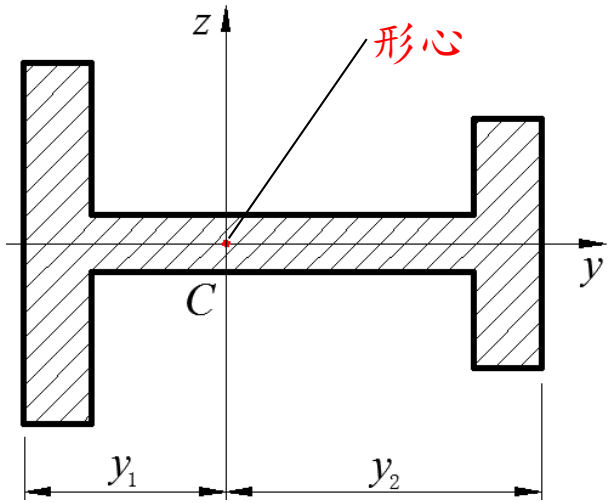
取危险点单元体—计算主应力

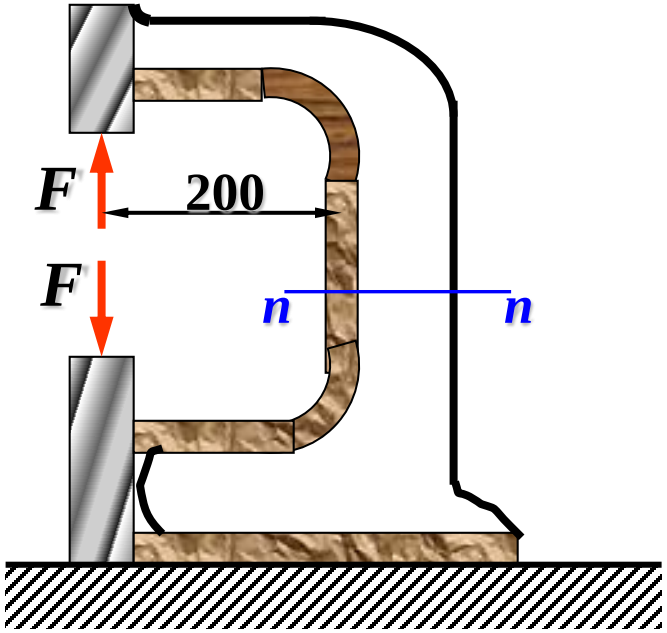
↓

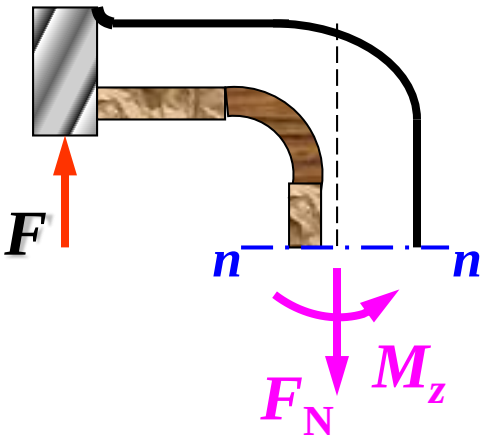
选择适当的强度理论校核

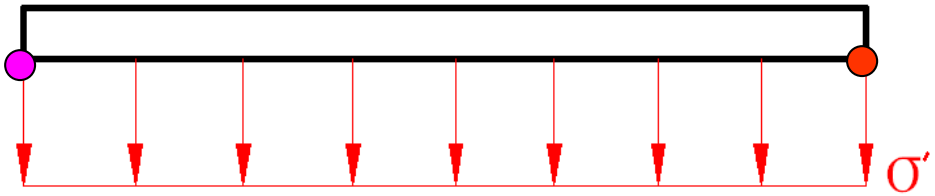
例题8-1 小型压力机的铸铁框架如图所示。已知材料的许用拉应力 $[\sigma_t]$ =30MPa，许用压应力 $[\sigma_c]$ =80MPa。F=12kN。截面尺寸：y_1=4.05cm，y_2=5.95cm，A=42cm^2，I_z=487.9cm^4。校核框架立柱的强度。

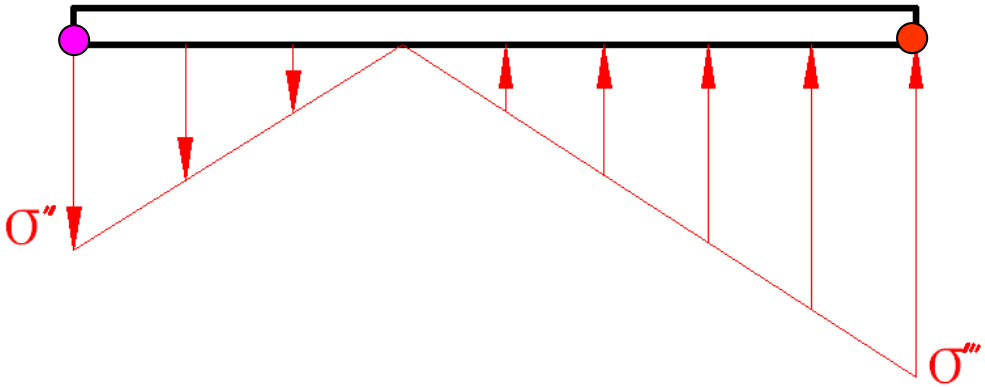








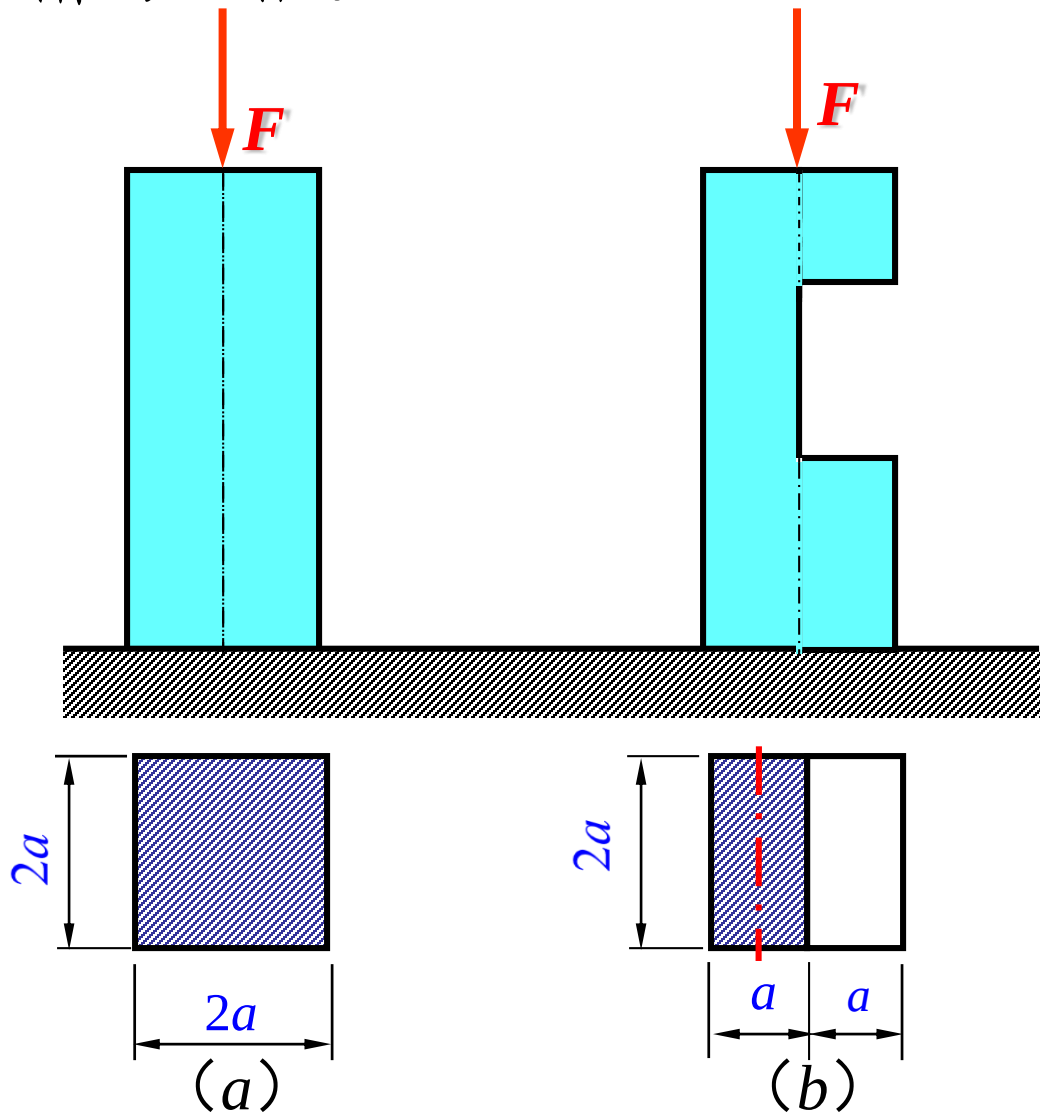




$$\sigma_t = 26.83\text{MPa}$$

$$\sigma_c = 32.34\text{MPa}$$

例题8-2 正方形截面立柱的中间处开一个槽，使截面面积为原来截面面积的一半。求开槽后立柱的的最大压应力是原来不开槽的几倍。



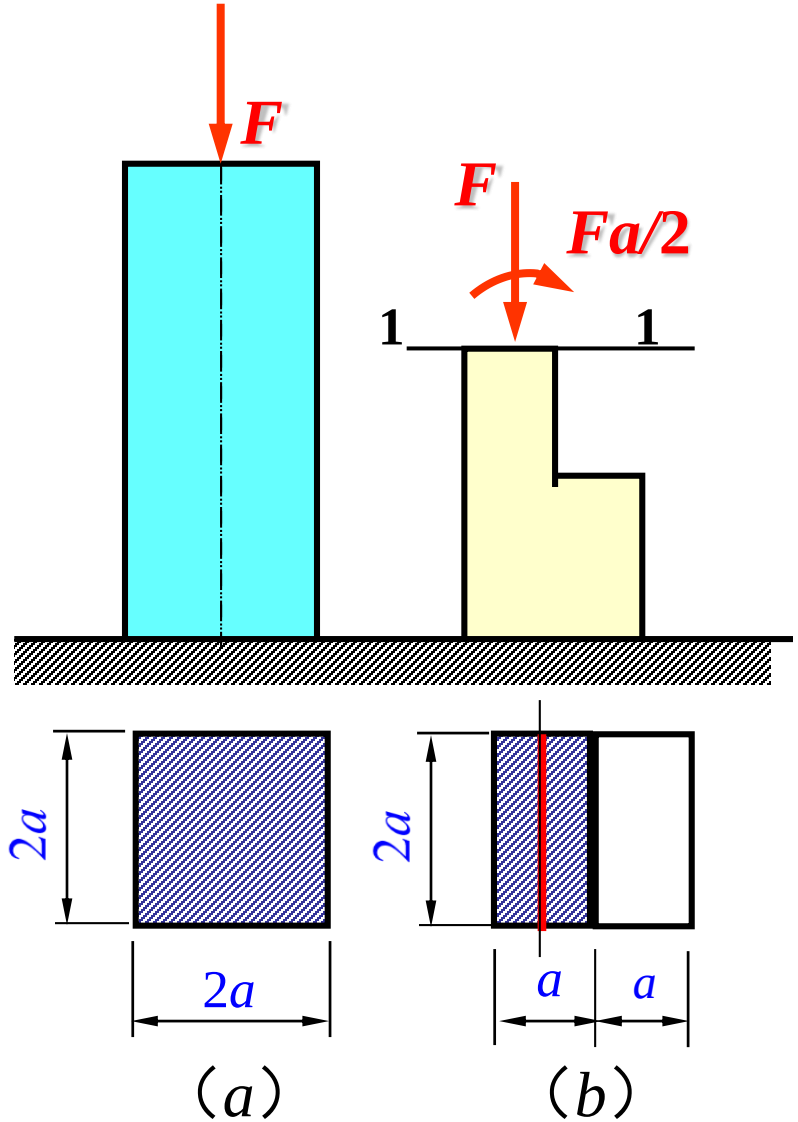
解：

$$\sigma_a = \frac{F_N}{A} = \frac{F}{A} = \frac{F}{(2a)^2} = \frac{F}{4a^2}$$

$$\sigma_b = \frac{F_N}{A} + \frac{M}{W} = \frac{F}{2a \cdot a} + \frac{Fa/2}{\frac{1}{6}2a \cdot a^2} = \frac{2F}{a^2}$$

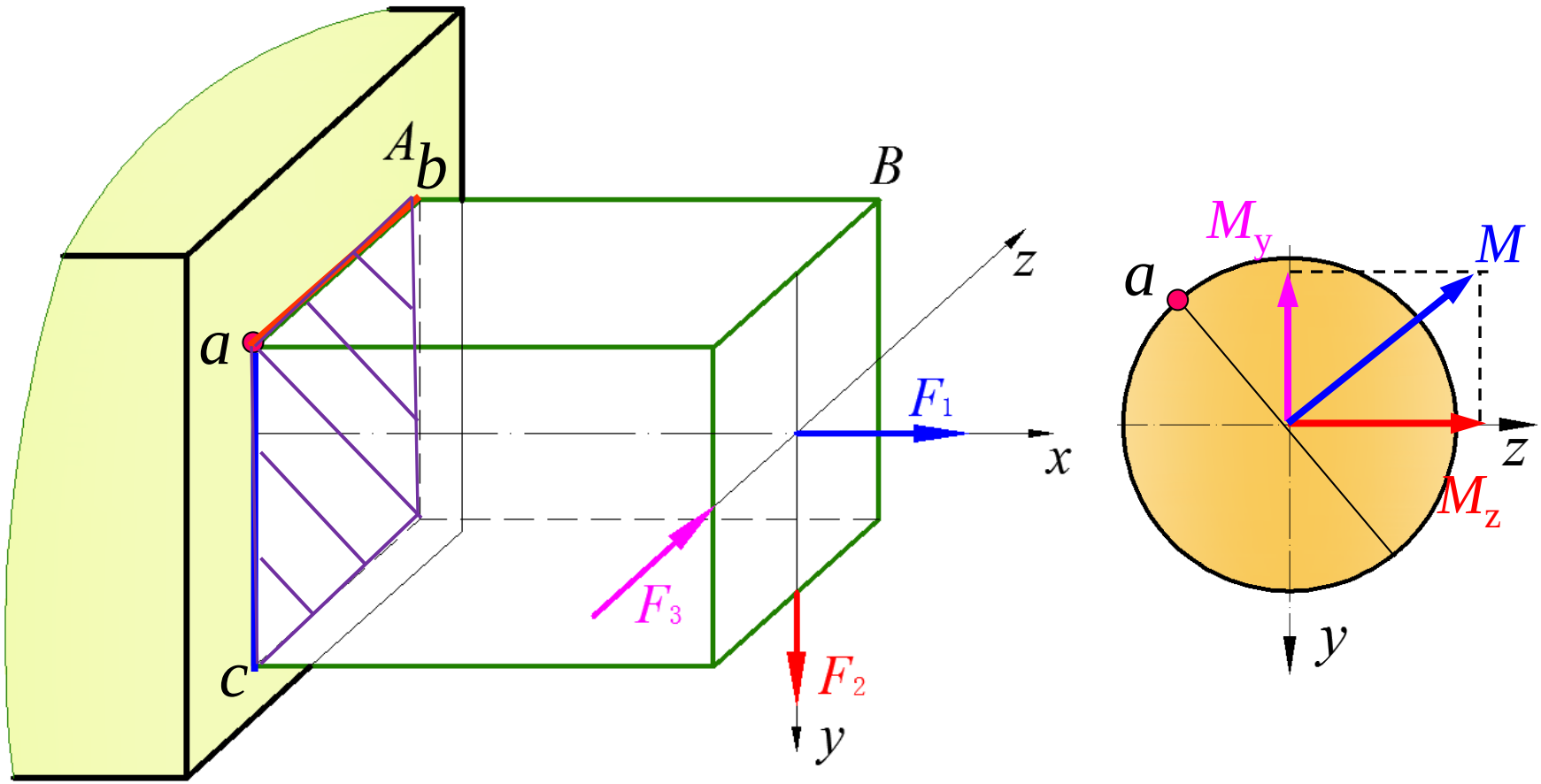
$$\frac{\text{开槽后立柱的最大压应力}}{\text{未开槽前立柱的最大压应力}}$$

$$= \frac{2F/a^2}{F/4a^2} = 8$$



讨论题

矩形截面构件，材料为铸铁，受力如图，F_1沿轴线，F_2、F_3分别沿对称轴y、z，试分析危险点的位置。



解组合变形的一般步骤

外力分析—绘受力简图

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内力分析—找危险面

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应力分析—找危险点

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取危险点单元体—计算主应力

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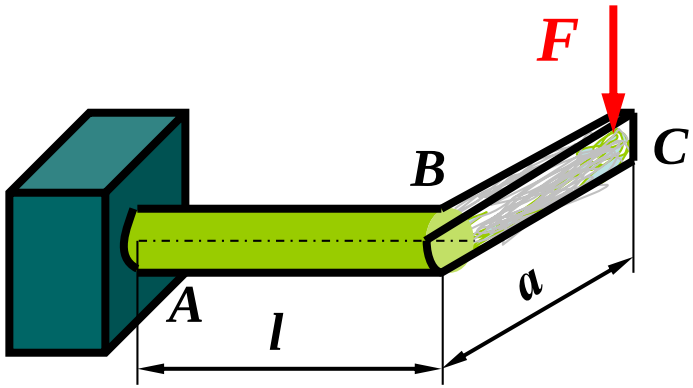
选择适当的强度理论校核

§8-4 扭转与弯曲的组合

（Combined bending and torsion）

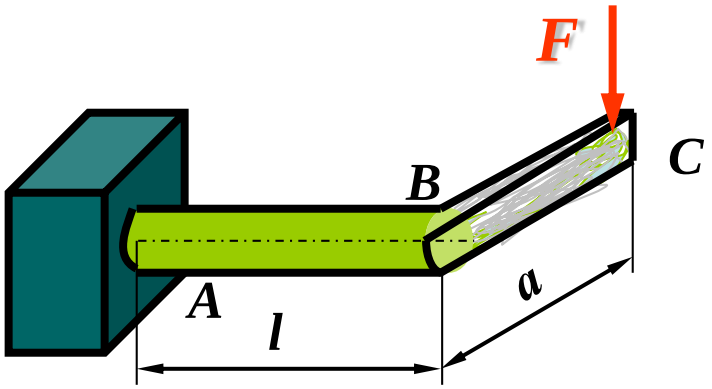


一直径为d的等直圆杆AB，B端具有与AB成直角的刚臂。

研究AB。

一、外力分析

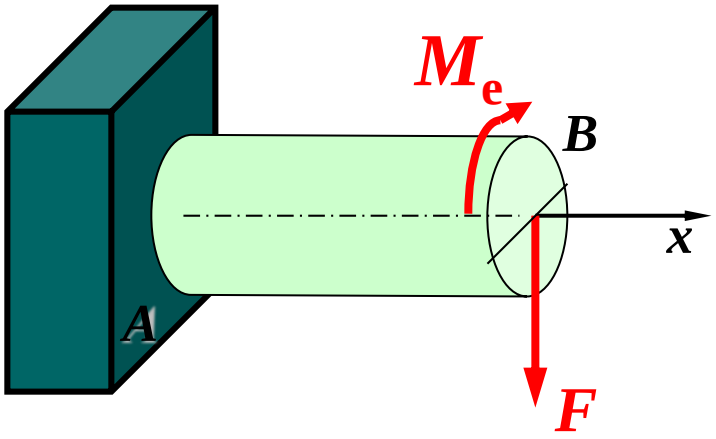
将力F向 AB 杆右端截面的形心B简化。



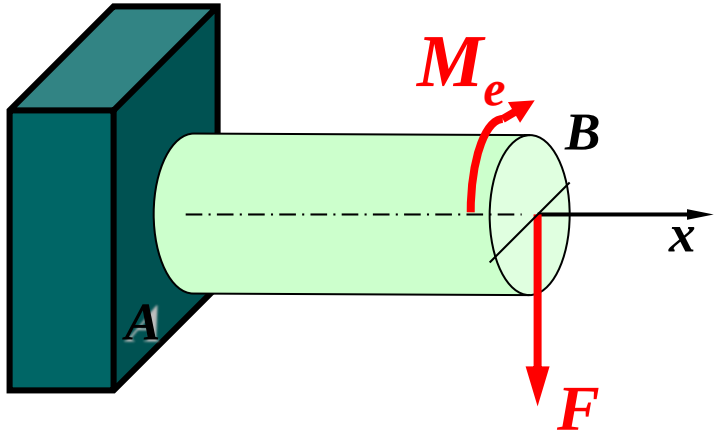
横向力F，引起弯曲变形。

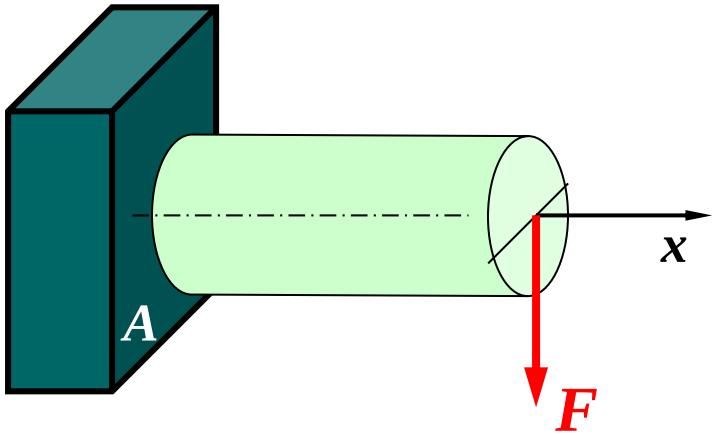
力偶矩 $M_e = Fa$，引起扭转变形。

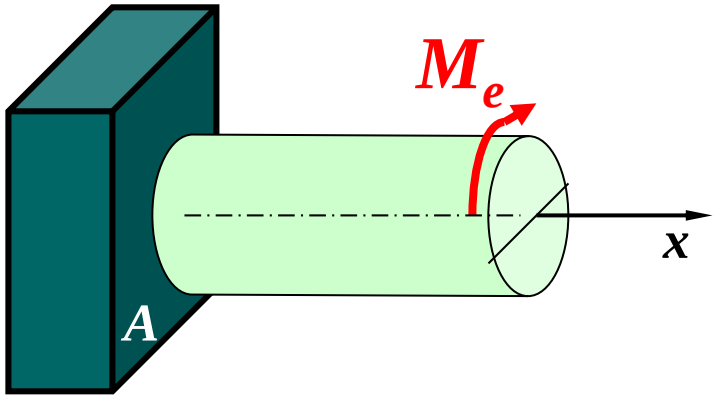
AB 杆为**弯曲与扭转组合变形**。

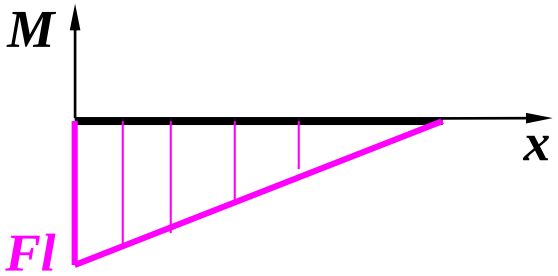


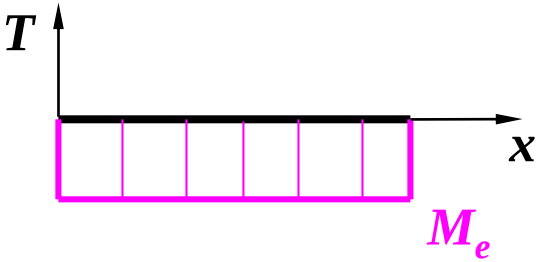
二、内力分析







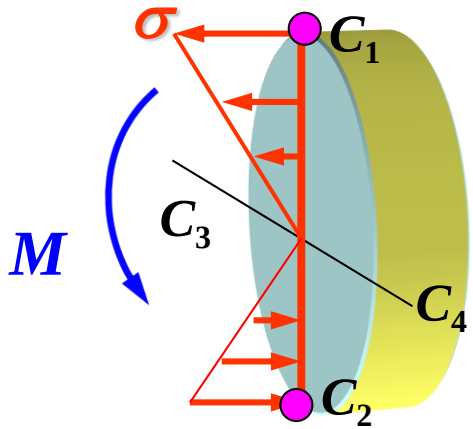


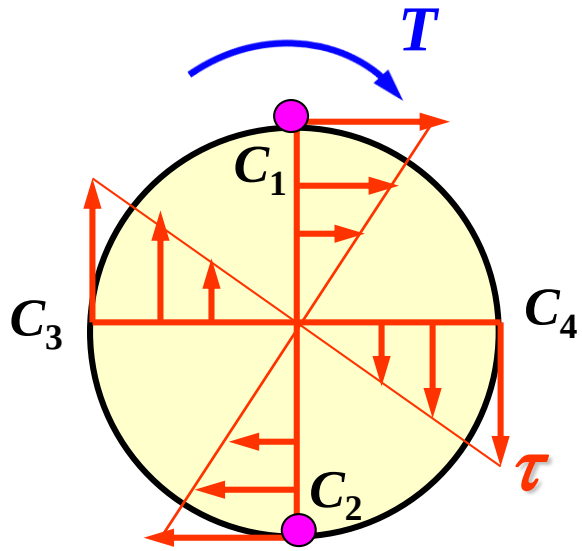


固定端A截面为**危险截面**。

三、应力分析

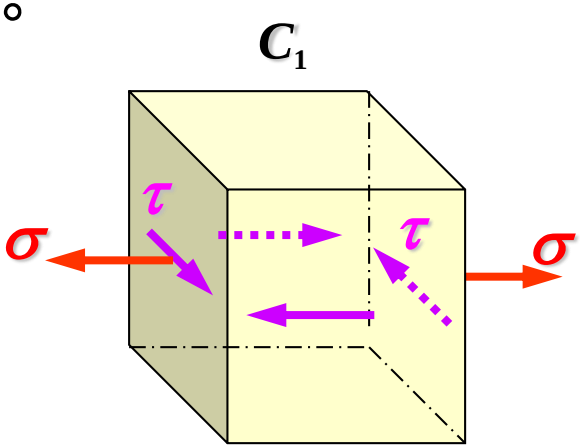
A截面





危险截面上的危险点为C_1和C_2点。

对于许用拉压应力相等的塑性材料制成的杆，这两点的危险程度是相同的，可取任意点C_1来研究。

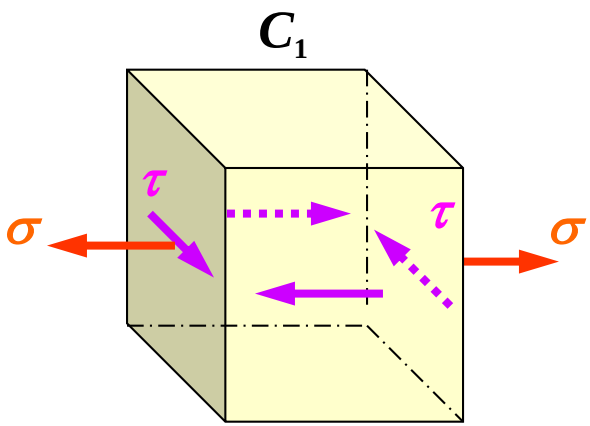


C_1点处于二向应力状态，该点的单元体如图示。

$$\sigma = \frac{M}{W_z} \qquad \tau = \frac{T}{W_t}$$

四、强度分析

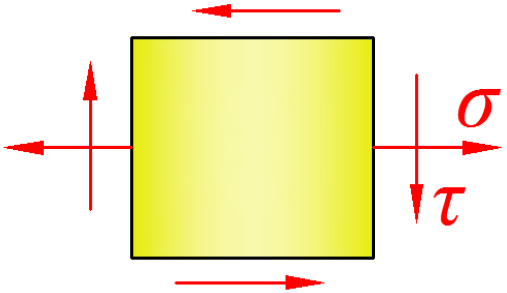
1、主应力计算



$$\begin{matrix}\sigma_{max}\\ \sigma_{min}\end{matrix}=\frac{\sigma}{2}\pm\sqrt{(\frac{\sigma}{2})^2+\tau^2}=\frac{\sigma}{2}\pm\frac{1}{2}\sqrt{\sigma^2+4\tau^2}$$

$$\sigma_1=\frac{\sigma}{2}+\frac{1}{2}\sqrt{\sigma^2+4\tau^2}\quad \sigma_2=0\quad \sigma_3=\frac{\sigma}{2}-\frac{1}{2}\sqrt{\sigma^2+4\tau^2}$$

2、相当应力计算



$$\sigma_1-\sigma_3\le[\sigma]$$

$$\sigma_{r3}=\sigma_1-\sigma_3=\sqrt{\sigma^2+4\tau^2}$$

$$\sqrt{\frac{1}{2}[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2]}\le[\sigma]$$

$$\sigma_{r4}=\sqrt{\sigma^2+3\tau^2}$$

3、强度校核

$$\sigma_r\le[\sigma]$$

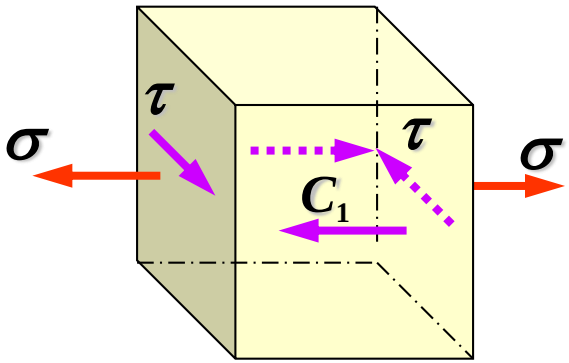
讨 论

（1） $\sigma_{r3}=\sigma_1-\sigma_3$ $\sigma_{r4}=\sqrt{\frac{1}{2}\left[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_1-\sigma_3)^2\right]}$

没有限制。

（2） $\sigma_{r3}=\sqrt{\sigma^2+4\tau^2}$

$\sigma_{r4}=\sqrt{\sigma^2+3\tau^2}$

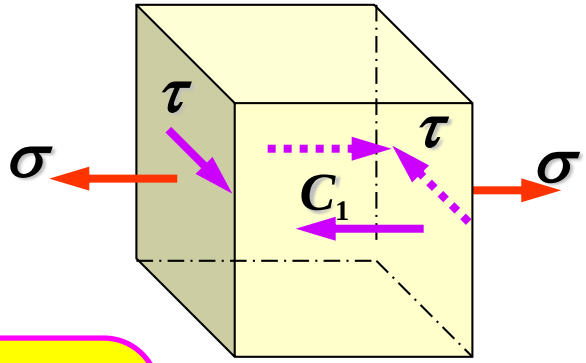


该公式适用于图示的平面应力状态，σ是危险点的正应力，τ是危险点的切应力。

适用：弯扭组合变形；拉（压）与扭转的组合变形；以及拉（压）扭转与弯曲的组合变形。

（3）对于圆形截面杆有

$$W_z = \frac{1}{2}W_t = \frac{\pi d^3}{32}$$



$$\sigma_{r3} = \sqrt{(\frac{M}{W_z})^2 + 4(\frac{T}{W_t})^2} = \frac{\sqrt{M^2 + T^2}}{W_z}$$

$$\sigma_{r4} = \sqrt{(\frac{M}{W_z})^2 + 3(\frac{T}{W_t})^2} = \frac{\sqrt{M^2 + 0.75T^2}}{W_z}$$

式中W_z为杆的**抗弯截面系数**；M，T分别为**危险截面**（同一截面）的弯矩和扭矩；以上两式只适用于**弯扭组合变形下的圆截面杆**。

$$\sigma_{r3} = \sigma_1 - \sigma_3$$

$$\sigma_{r4} = \sqrt{\frac{1}{2}\left[\left(\sigma_1 - \sigma_2\right)^2 + \left(\sigma_1 - \sigma_3\right)^2 + \left(\sigma_2 - \sigma_3\right)^2\right]}$$

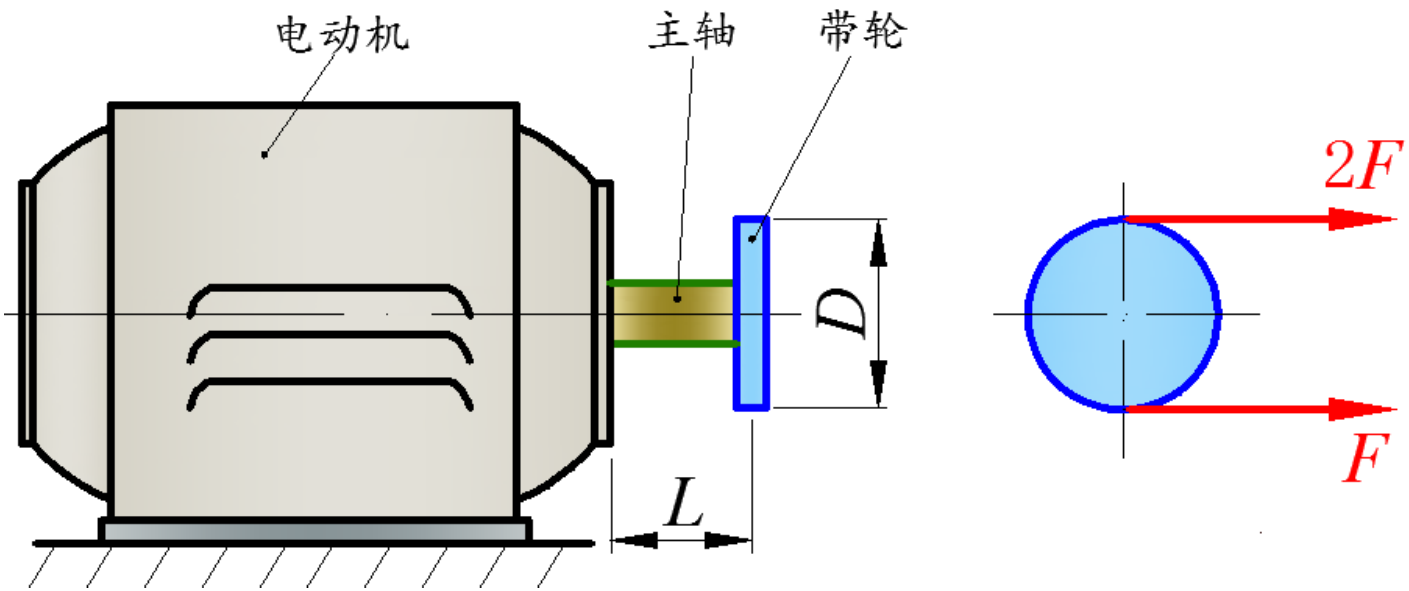
$$\sigma_{r3} = \sqrt{\sigma^2 + 4\tau^2}$$

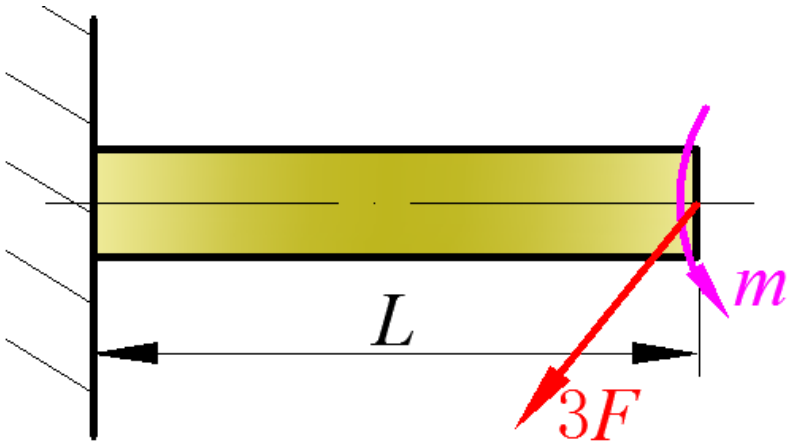
$$\sigma_{r4} = \sqrt{\sigma^2 + 3\tau^2}$$

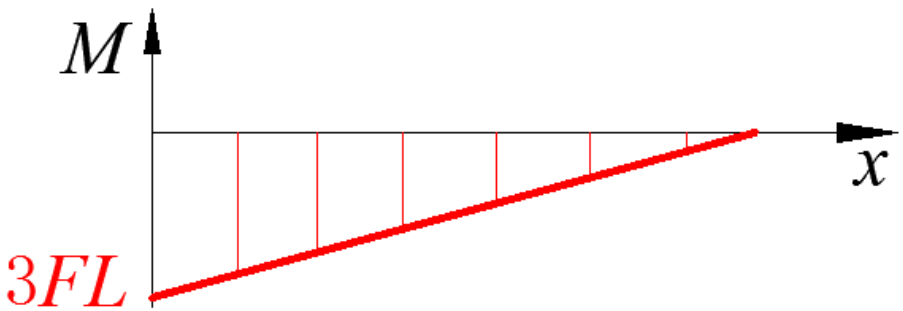
$$\sigma_{r3} = \frac{\sqrt{M^2 + T^2}}{W_Z}$$

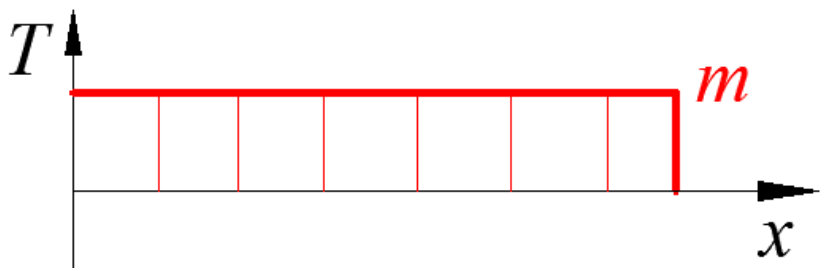
$$\sigma_{r4} = \frac{\sqrt{M^2 + 0.75T^2}}{W_Z}$$

例题8-3 电动机功率P=9kW，转速n=715r/min，带轮直径D=250mm，主轴外伸部分长L=120mm，$[\sigma]$=60MPa。按第三、第四强度理论分别估算主轴直径。







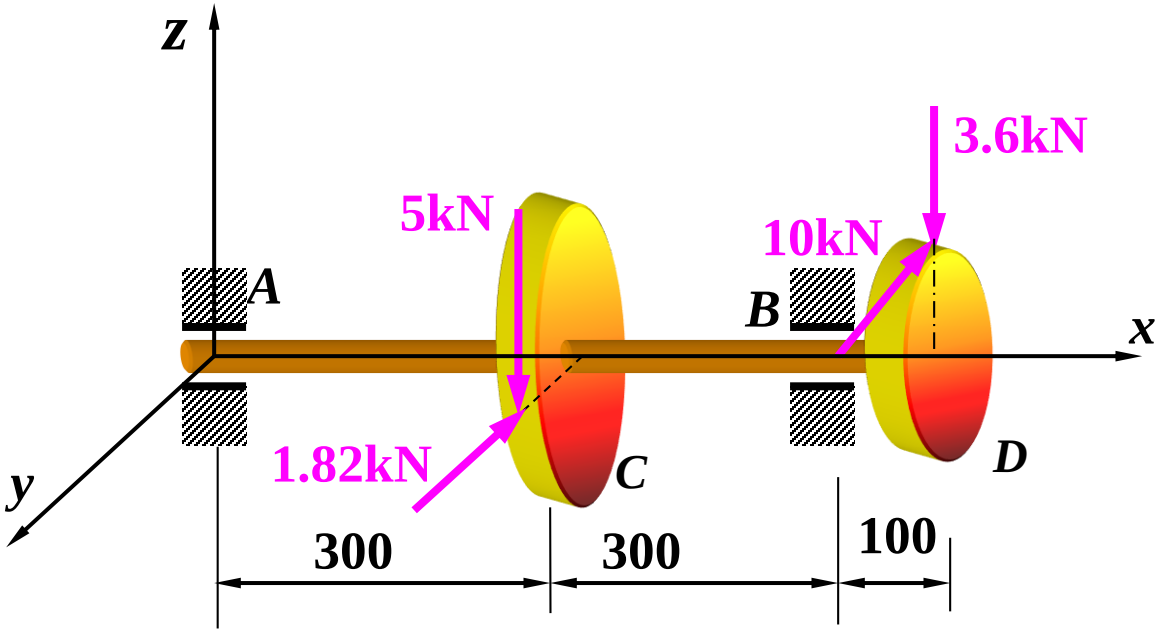


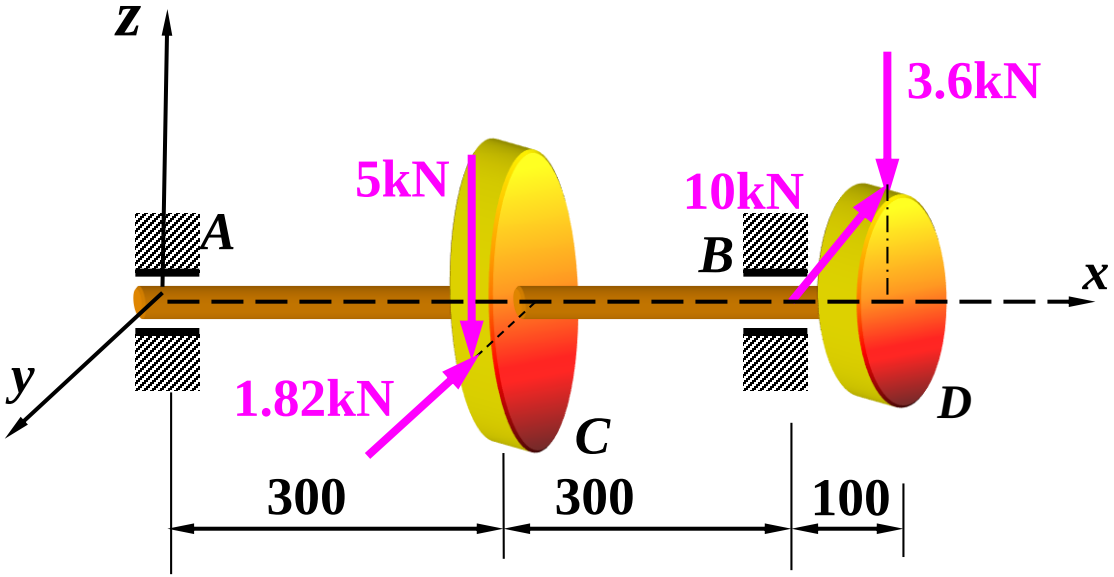
第三：$d \geq 0.0396\text{m}$

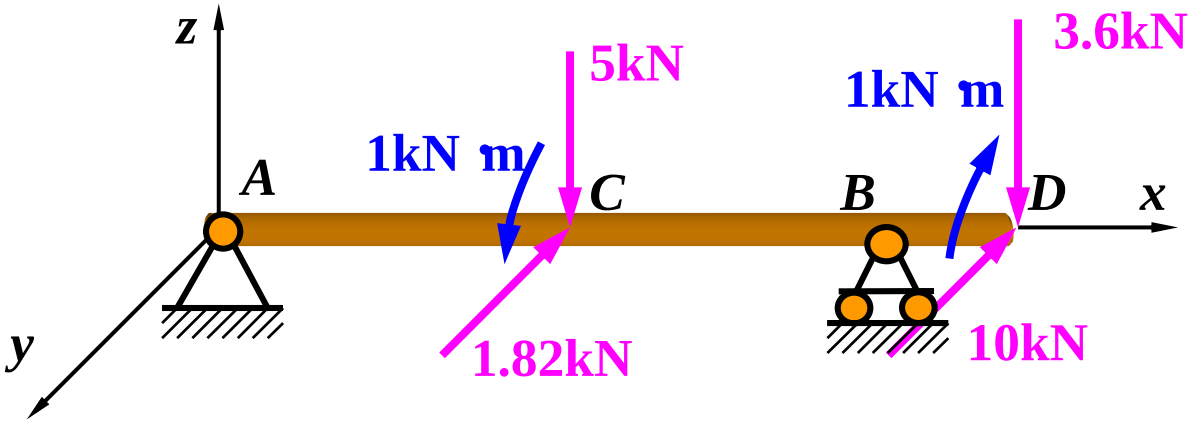
第四：$d \geq 0.0394\text{m}$

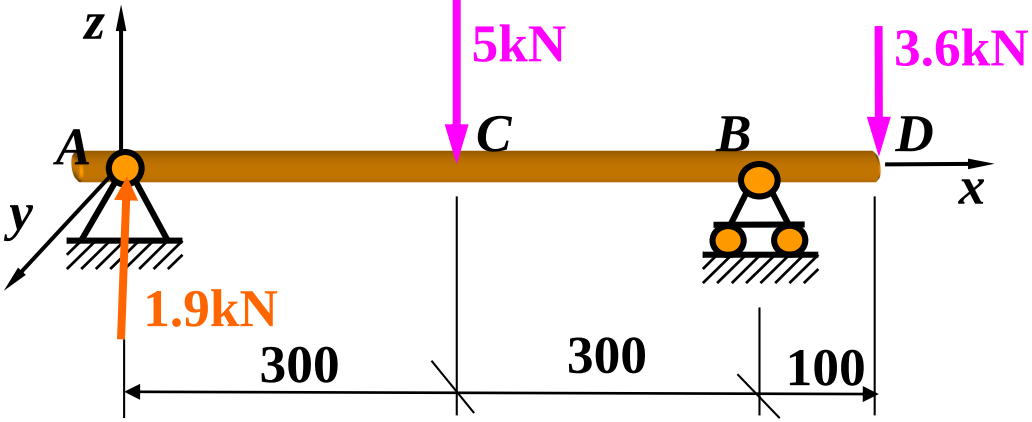
例题8-4 图示一钢制实心圆轴，轴上的齿轮C上作用有铅垂切向力5kN，径向力1.82 kN；齿轮D上作用有水平切向力10 kN，径向力3.6 kN。圆轴的直径d=55mm，许用应力 [σ]=100 MPa。齿轮C的节圆直径 d_1 = 400 mm，齿轮D的节圆直径d_2=200mm。

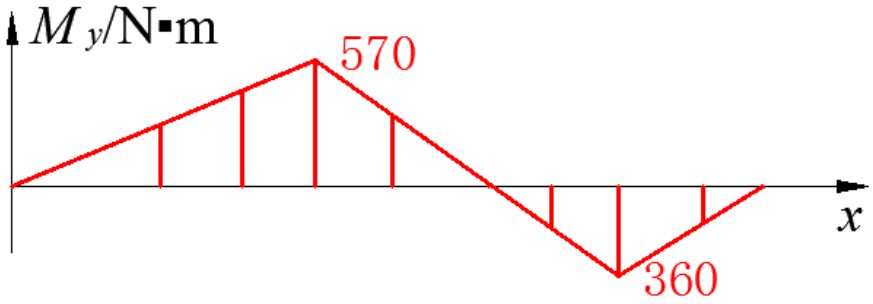
求：（1）确定危险点的位置，用单元体表示危险点的应力状态，并求主应力和最大切应力；（2）用第三强度理论校核轴的强度。

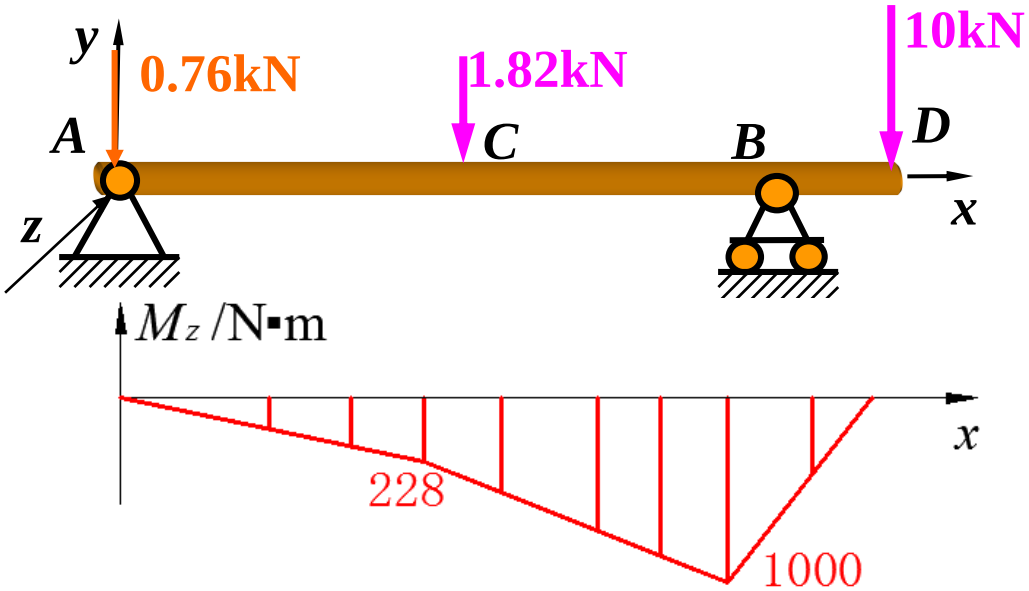



z
3.6kN
5kN
10kN
A
B
x
y
1.82kN
C
D
300
300
100


z
3.6kN
5kN
1kN m
1kN m
A
C
B
D
x
y
1.82kN
10kN

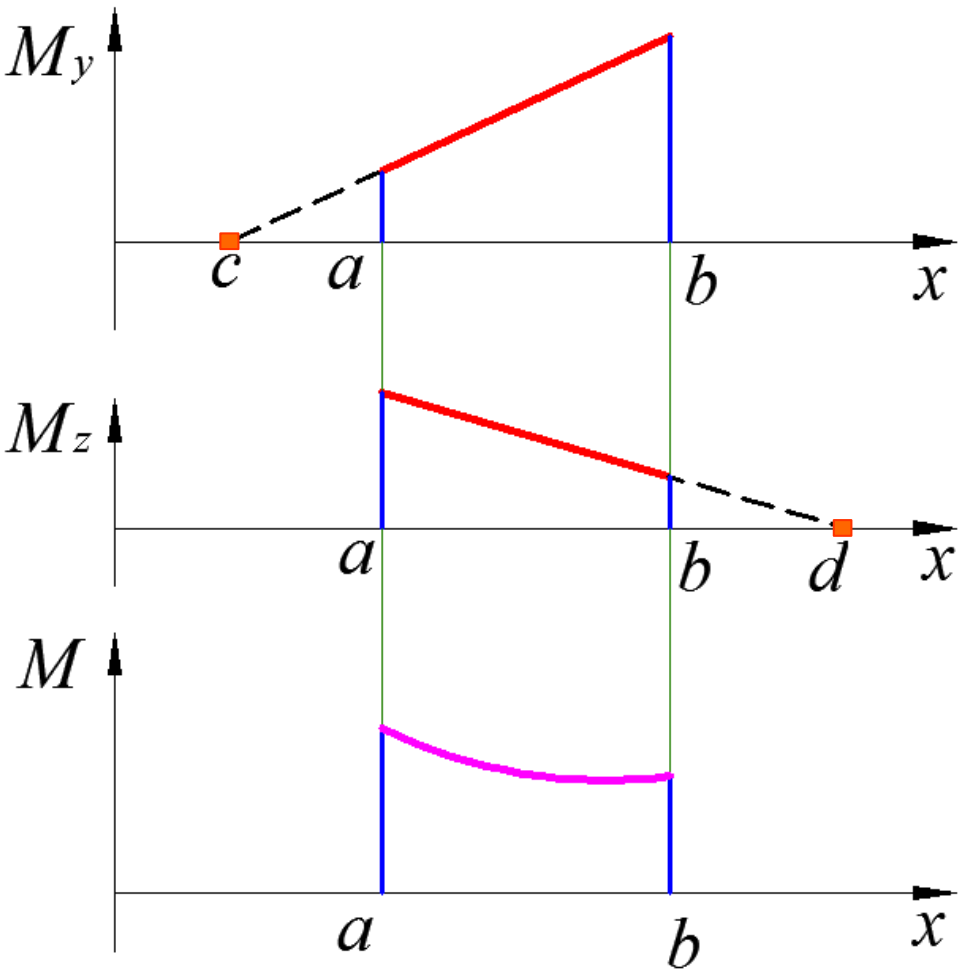

z
5kN
3.6kN
A
C
B
D
x
y
1.9kN
300
300
100

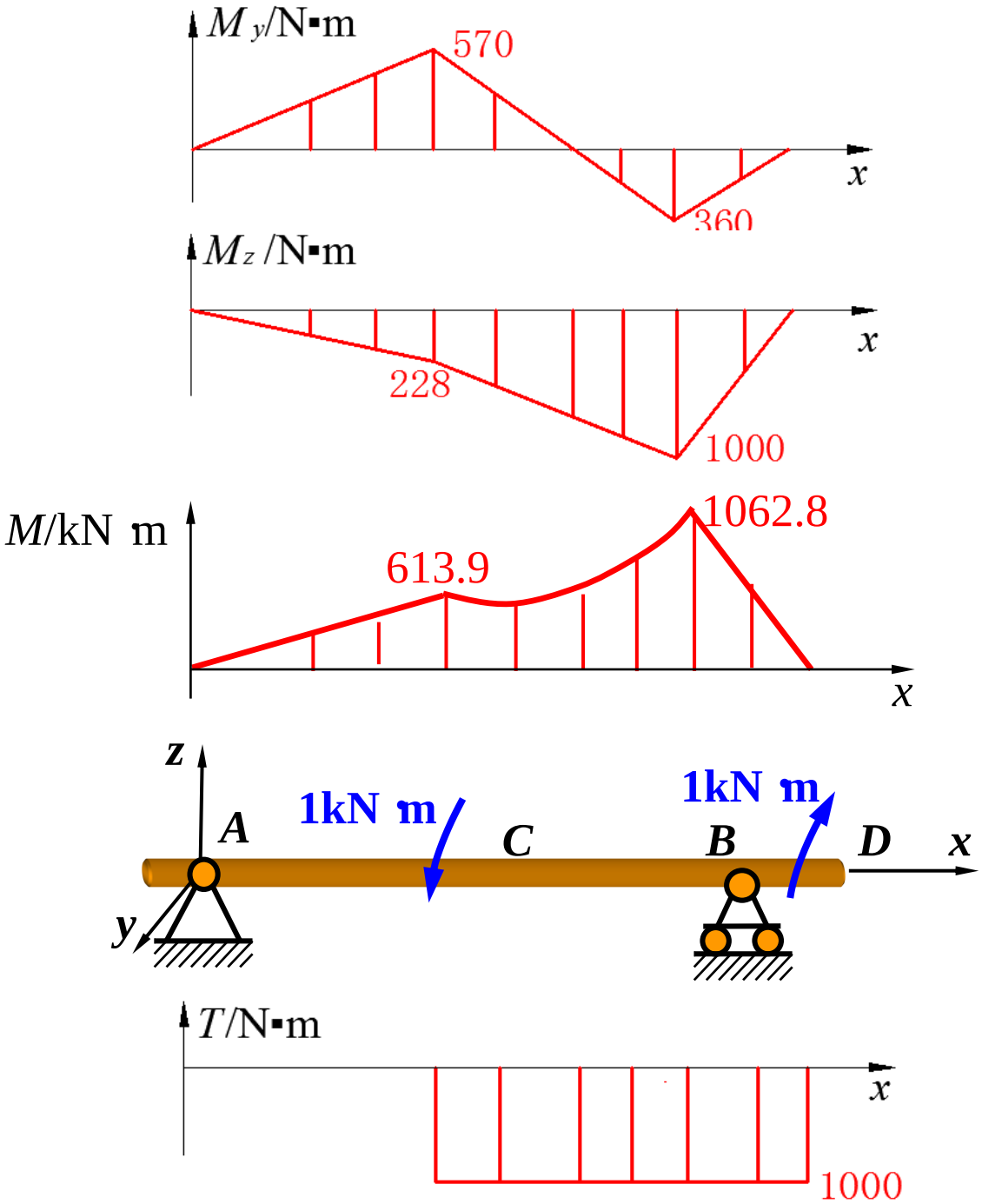

M_y/N▪m
570
x
360

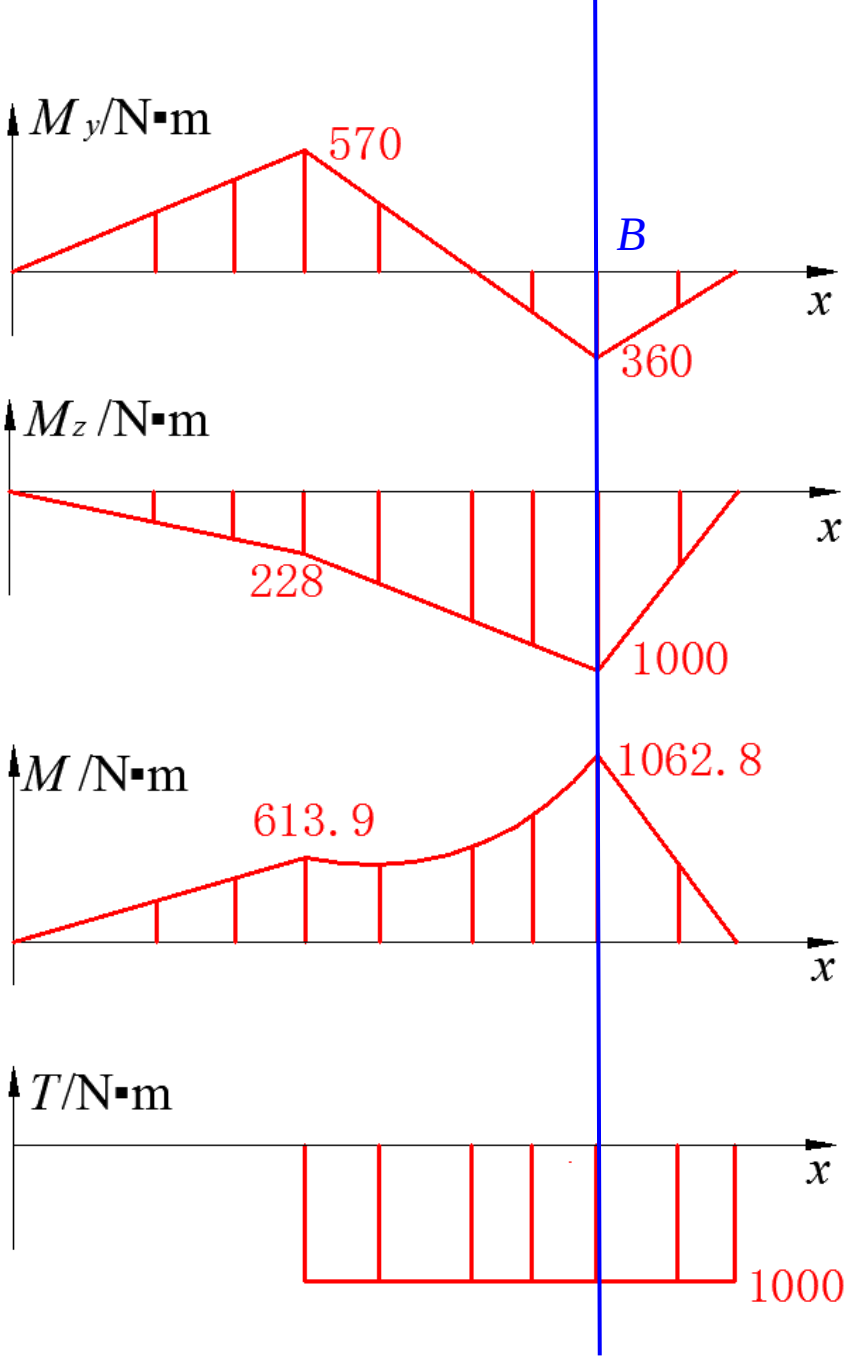

y
0.76kN
1.82kN
10kN
A
C
B
D
x
z
M_z /N▪m
x
228
1000

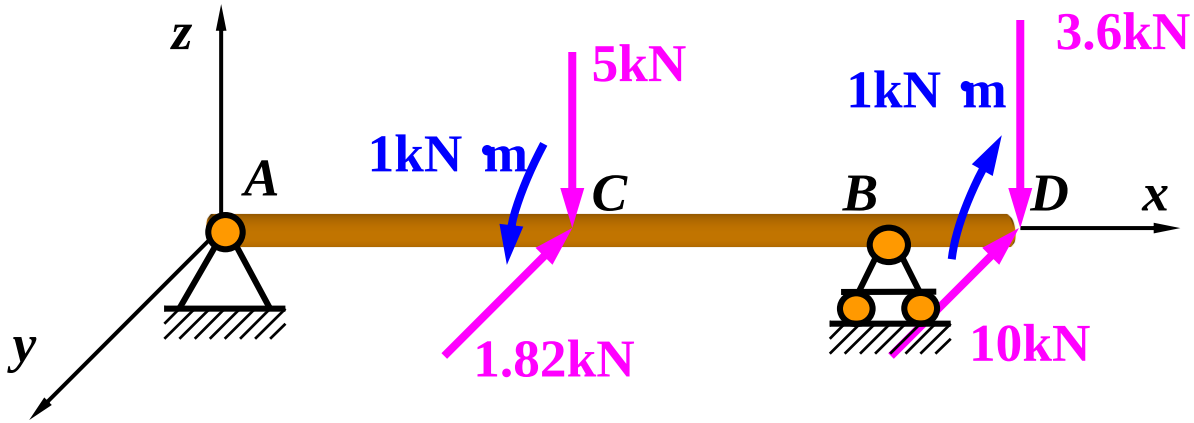
某段梁的弯矩M_y和M_z图如图所示，它们均为直线，且其延长线分别与x轴交于c点、d点。如果c点和d点不重合，则该段总弯矩必为**凹曲线**。

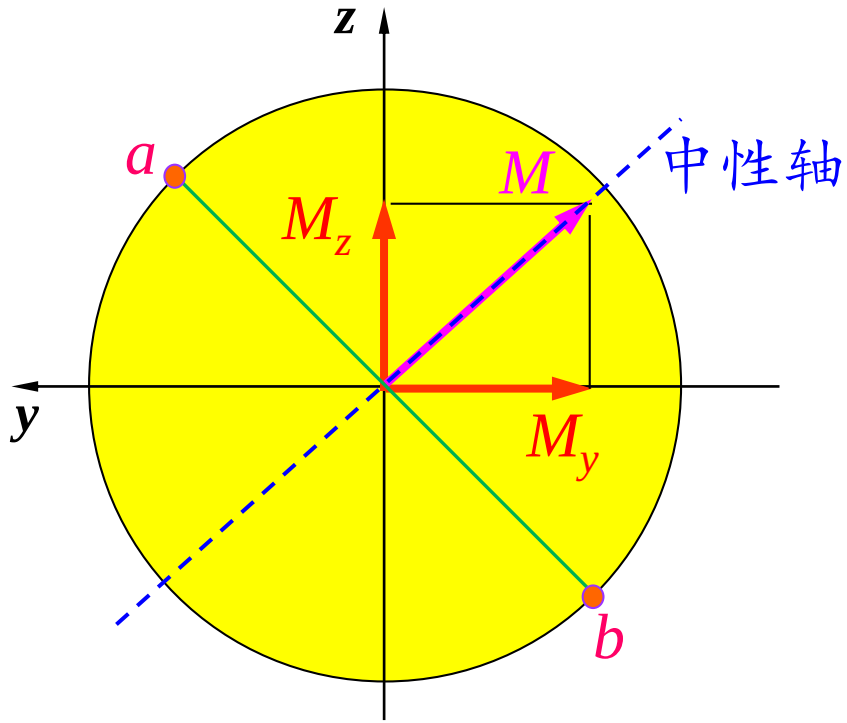
$$M=\sqrt{M_y^2+M_z^2}$$

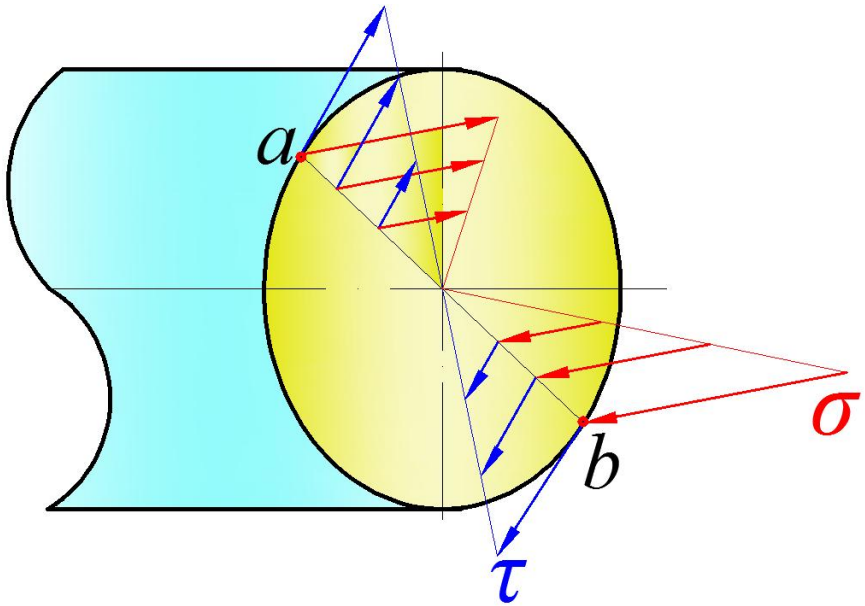


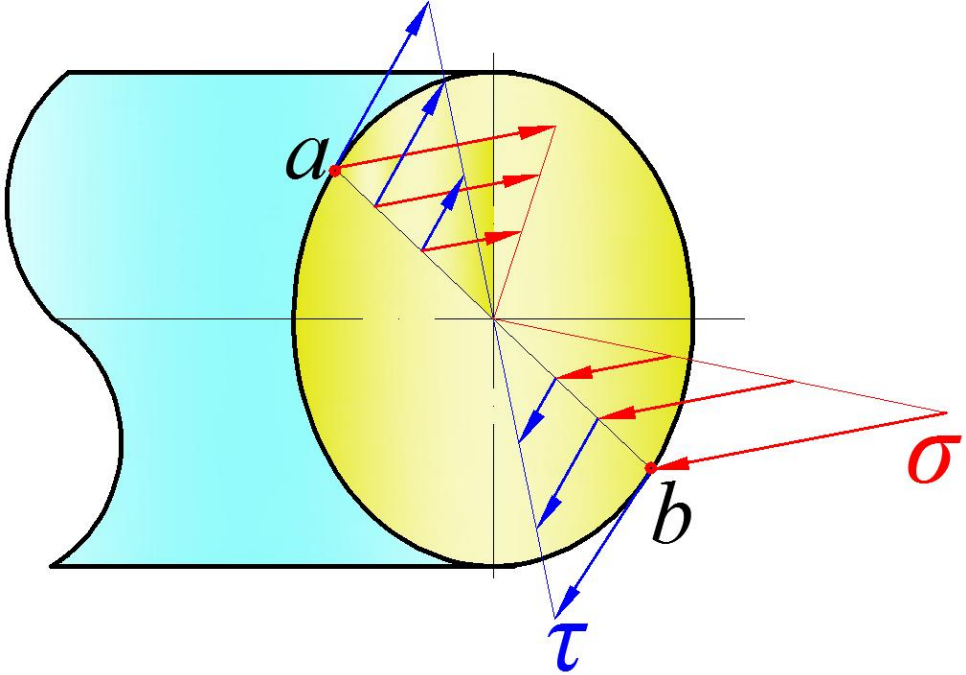

M_y/N▪m
570
360
x
M_z /N▪m
228
1000
x
M/kN m
613.9
1062.8
x
z
A
1kN m
C
1kN m
B
D
x
y
T/N▪m
x
1000

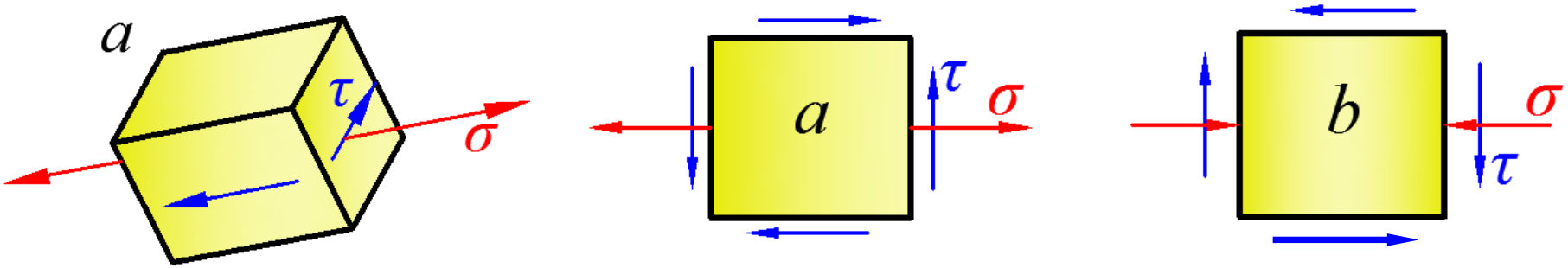

M_y/N▪m
570
B
x
360
M_z /N▪m
x
228
1000
M /N▪m
1062. 8
613. 9
x
T/N▪m
x
1000


z
5kN
3.6kN
1kN m
1kN m
A
C
B
D
x
y
1.82kN
10kN


z
a
M
中性轴
M_z
y
M_y
b


a
σ
b
τ

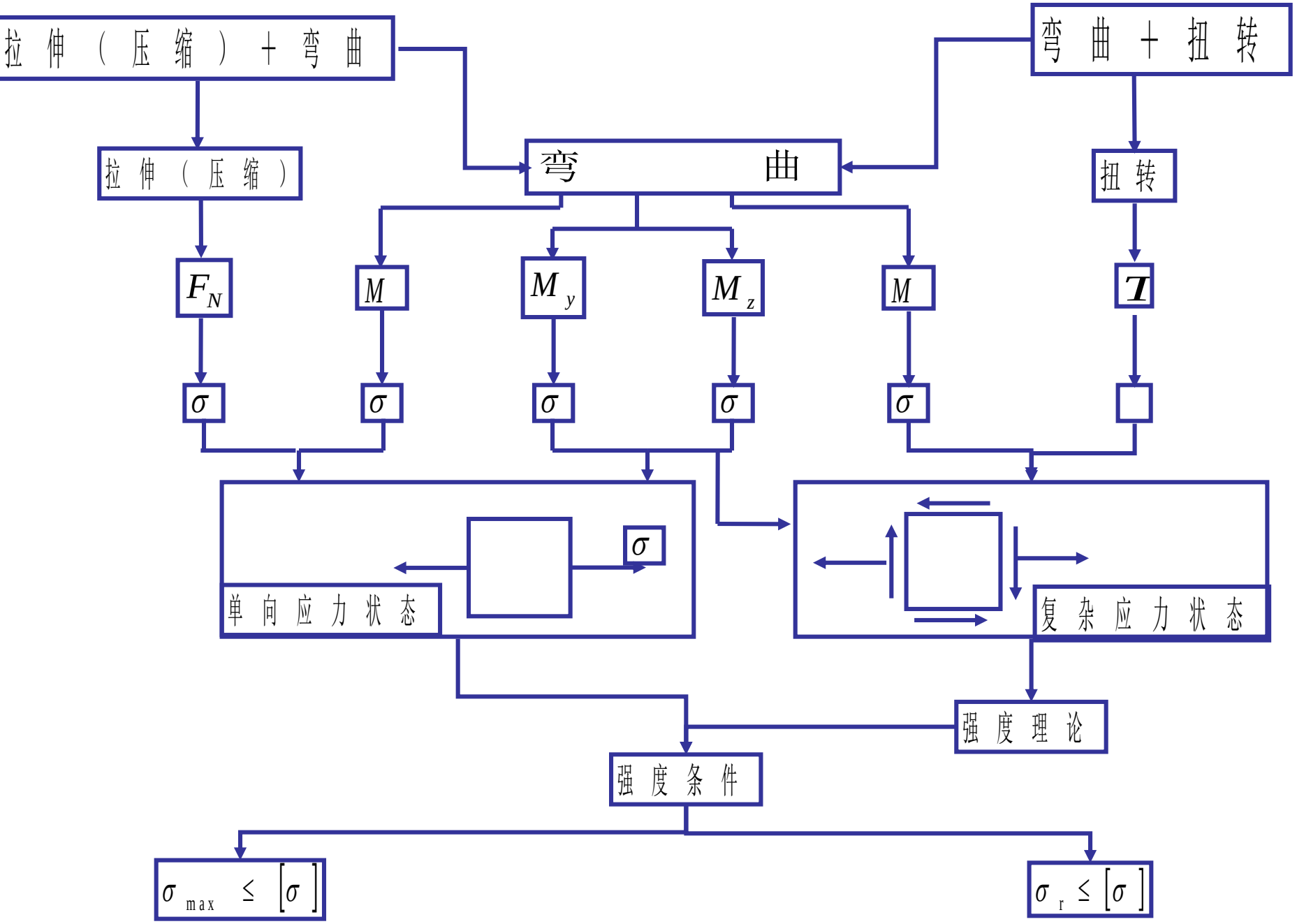


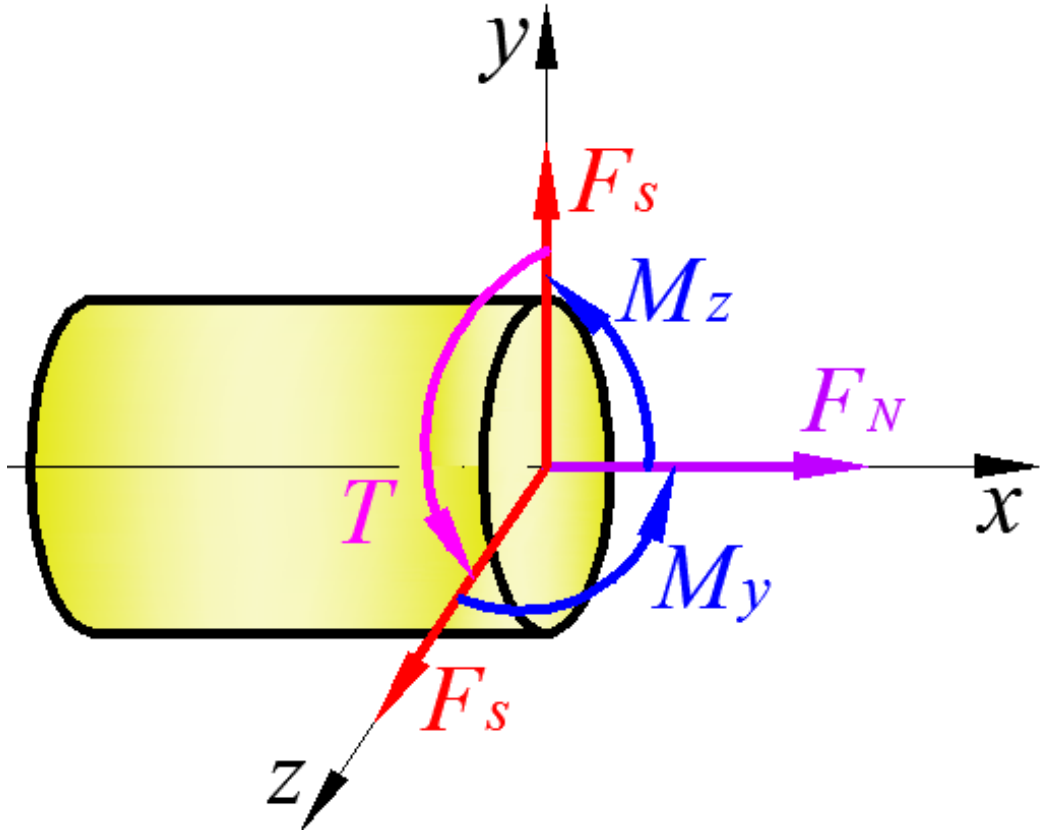


σ =65.07MPa， τ =30.61MPa

σ_1=77.21MPa， σ_2=0， σ_3=-12.14MPa

σ_{r3}=89.35MPa， τ_{max}=44.67MPa


拉 伸 （ 压 缩 ） + 弯 曲
弯 曲 + 扭 转
拉 伸 （ 压 缩 ）
弯 曲
扭 转
F_N
M
M_y
M_z
M
T
σ
σ
σ
σ
σ
σ
单 向 应 力 状 态
复 杂 应 力 状 态
强 度 理 论
强 度 条 件
$\sigma_{max} \le [\sigma]$
$\sigma_r \le [\sigma]$



$$\left.\begin{array}{l} F_N \\ M_y \\ M_z \end{array}\right\} \longrightarrow \sigma$$

$$\left.\begin{array}{l} T \\ F_s \end{array}\right\} \longrightarrow \tau$$

第八章结束